Chapter 6

Campbell Biology, AP Edition, 12e

A microscopic view of a Paramecium shows various structures within the single-celled organism.

Chapter 6

A Tour of the Cell

Figure 6.1

The cell is an organism's basic unit of structure and function. Many forms of life exist as single-celled organisms, such as the *Paramecium* shown here. Larger, more complex organisms, including plants and animals, are multicellular. In this chapter, we focus mainly on eukaryotic cells—cells with a nucleus.

How does the internal organization of eukaryotic cells allow them to perform the functions of life?

Energy and matter transformations: A system of internal membranes synthesizes and modifies proteins, lipids, and carbohydrates; chloroplasts convert light energy to chemical energy; mitochondria break down molecules, generating A T P.

Genetic information storage and transmission: D N A in the nucleus contains instructions for making proteins; ribosomes are the sites of protein synthesis.

Interactions with the environment: The plasma membrane controls what goes into and out of the cell; plant cells have a protective cell wall.

Key Concepts

- **6.1Biologists use microscopes and biochemistry to study cells****
- **6.2Eukaryotic cells have internal membranes that compartmentalize their functions****
- **6.3The eukaryotic cell's genetic instructions are housed in the nucleus and carried out by the ribosomes****
- **6.4The endomembrane system regulates protein traffic and performs metabolic functions****

- **6.5Mitochondria and chloroplasts change energy from one form to another****
- **6.6The cytoskeleton is a network of fibers that organizes structures and activities in the cell****
- **6.7Extracellular components and connections between cells help coordinate cellular activities****
- **6.8A cell is greater than the sum of its parts****

Advanced placement edition

Many structures and processes of cells that maintain dynamic homeostasis **(BIG IDEA 2)** and respond to the environment **(BIG IDEA 3)** reflect a shared ancestry **(BIG IDEA 1).**

Study Tip

Draw animal and plant cells: Draw an outline of an animal cell and add structures, labels, and functions. Draw a plant cell labeled with structures unique to plant cells.

A drawn outline is roughly a circle labeled plasma membrane a selective barrier.

Get Ready for This Chapter

In this chapter, you will learn about the organization, structure, and function of cells, including the process of gene expression. Before beginning this chapter, you should be able to do the following:

- Explain the role of emergent properties in biological systems (see Concept 1.1, section entitled "Theme: New Properties Emerge at Successive Levels of Biological Organization").
- Describe the structure of a phospholipid (see Concept 5.3, section entitled "Phospholipids").
- List the three domains of life and differentiate between prokaryotes and eukaryotes (see Concept 1.2, section entitled "The Three Domains of Life").
- Describe the process of gene expression (see Concept 1.1, section entitled "Theme: Life's Processes Involve the Expression and Transmission of Genetic Information").

The following questions provide a quick check of your basic knowledge in these areas.

1. Which of the following expresses the concept of emergent properties?

```
    A cell is the basic unit of structure in living organisms.
```

That's incorrect. Try again.

Incorrect. Review the following material: The study of life on Earth extends from the molecular scale to the microscopic scale of the cells that make up organisms to the global scale of the entire living planet. We can divide this enormous range into different levels of biological organization. Starting at the molecular level and zooming out, novel properties emerge at each level that are absent from the preceding one. These emergent properties are due to the arrangement and interactions of parts as complexity increases.

```
        The parts of a car, once assembled, result in a
structure that can provide transportation.
```

That's correct.

Correct! If you had trouble with this question, review the following material: The study of life on Earth extends from the molecular scale to the microscopic scale of the cells that make up organisms to the global scale of the entire living planet. We can divide this enormous range into different levels of biological organization. Starting at the molecular level and zooming out, novel properties emerge at each level that are absent from the preceding one. These emergent properties are due to the arrangement and interactions of parts as complexity increases.

```
        Energy can be transformed but not created or destroyed,
while matter cycles through a system.
```

That's incorrect. Try again.

Incorrect. Review the following material: The study of life on Earth extends from the molecular scale to the microscopic scale of the cells that make up organisms to the global scale of the entire living planet. We can divide this enormous range into different levels of biological organization. Starting at the molecular level and zooming out, novel properties emerge at each level that are absent from the preceding one. These emergent properties are due to the arrangement and interactions of parts as complexity increases.

```
        Properties emerge from a system through evolutionary
mechanisms.
```

That's incorrect. Try again.

Incorrect. Review the following material: The study of life on Earth extends from the molecular scale to the microscopic scale of the cells that make up organisms to the global scale of the entire living planet. We can

divide this enormous range into different levels of biological organization. Starting at the molecular level and zooming out, novel properties emerge at each level that are absent from the preceding one. These emergent properties are due to the arrangement and interactions of parts as complexity increases.

2. Membrane phospholipids

```
     may have “kinks” in their tails, each caused by the
presence of a double bond between carbons.
```

That's correct.

Correct! If you had trouble with this question, review the following material: Phospholipids are arranged in a bilayer. Phospholipid tails are comprised of hydrocarbons, which are hydrophobic and excluded from water in the center of the bilayer. Phospholipid heads have a charged phosphate group and attachments that form a hydrophilic head that has affinity for water on the outside of the bilayer. The hydrocarbon tails may include double bonds, which each create a “kink” in the tail.

```
        remain fluid because they are tightly packed against one
another.
```

That's incorrect. Try again.

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```
        have hydrophilic tails that face outward and are exposed
to water and hydrophobic heads that face the center of the
membrane and are shielded from water.
```

That's incorrect. Try again.

Incorrect. Review the following material: Phospholipids are arranged in a bilayer. Phospholipid tails are comprised of hydrocarbons, which are hydrophobic and excluded from water in the center of the bilayer. Phospholipid heads have a charged phosphate group and attachments that form a hydrophilic head that has

affinity for water on the outside of the bilayer. The hydrocarbon tails may include double bonds, which each create a “kink” in the tail.

```
form impermeable layers for cells.
```

That's incorrect. Try again.

Incorrect. Review the following material: Phospholipids are arranged in a bilayer. Phospholipid tails are comprised of hydrocarbons, which are hydrophobic and excluded from water in the center of the bilayer. Phospholipid heads have a charged phosphate group and attachments that form a hydrophilic head that has affinity for water on the outside of the bilayer. The hydrocarbon tails may include double bonds, which each create a “kink” in the tail.

3. One difference between eukaryotic and prokaryotic cells is that eukaryotic cells

```
have membrane-enclosed organelles, which are lacking in prokaryotic cells.
```

That's correct.

Correct! If you had trouble with this question, review the following material: All cells share certain characteristics that allow cells to perform all activities required for life. All cells have both a membrane (the plasma membrane) that acts as a selective barrier and DNA, which is the cell’s genetic material. Eukaryotic cells have added complexity with extensive, elaborately arranged internal membranes that divide the cell into compartments called organelles.

```
are enclosed by a membrane, which is lacking in prokaryotic cells.
```

That's incorrect. Try again.

Incorrect. Review the following material: All cells share certain characteristics that allow cells to perform all activities required for life. All cells have both a membrane (the plasma membrane) that acts as a selective barrier and DNA, which is the cell’s genetic material. Eukaryotic cells have added complexity with extensive, elaborately arranged internal membranes that divide the cell into compartments called organelles.

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```
lack the complexity seen in prokaryotic cells.
```

That's incorrect. Try again.

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4. Which of the following options best depicts the flow of information when a gene directs the synthesis of a cellular component?

```
RNA to DNA to RNA to protein
```

That's incorrect. Try again.

Incorrect. Review the following material: The entire process by which the information in a gene directs the manufacture of a cellular product is called gene expression. The sequence of DNA nucleotides along a gene is transcribed, forming an RNA that is then translated. This results in a linked series of protein subunits called amino acids. An amino acid chain, a polypeptide, folds into a specific protein with a unique shape and function.

```
DNA to amino acid to RNA to protein
```

That's incorrect. Try again.

Incorrect. Review the following material: The entire process by which the information in a gene directs the manufacture of a cellular product is called gene expression. The sequence of DNA nucleotides along a gene is transcribed, forming an RNA that is then translated. This results in a linked series of protein subunits called amino acids. An amino acid chain, a polypeptide, folds into a specific protein with a unique shape and function.

```
DNA to RNA to protein
```

That's correct.

Correct! If you had trouble with this question, review the following material: The entire process by which the information in a gene directs the manufacture of a cellular product is called gene expression. The sequence of DNA nucleotides along a gene is transcribed, forming an RNA that is then translated. This results in a linked series of protein subunits called amino acids. An amino acid chain, a polypeptide, folds into a specific protein with a unique shape and function.

```
protein to RNA to DNA
```

That's incorrect. Try again.

Incorrect. Review the following material: The entire process by which the information in a gene directs the manufacture of a cellular product is called gene expression. The sequence of DNA nucleotides along a gene is transcribed, forming an RNA that is then translated. This results in a linked series of protein subunits called amino acids. An amino acid chain, a polypeptide, folds into a specific protein with a unique shape and function.

Learning Objectives

After studying this chapter, you should be able to do the following:

- 6.1Identify how common techniques of microscopy and biochemistry are used in cell biology.
- 6.2Compare and contrast (1) prokaryotic and eukaryotic cells and (2) animal and plant cells.
- 6.3Describe the structure of the nucleus, chromosomes, and ribosomes, and explain their functions.
- 6.4Identify the parts of the endomembrane system and describe their roles in the cell.
- 6.5Compare and contrast the structures and functions of mitochondria and chloroplasts.
- 6.6Describe the subunits and structures of the three types of cytoskeletal fibers and list their functions.
- 6.7Compare and contrast (1) the extracellular components of plant and animal cells and (2) the cell junctions of plant and animal cells.
- 6.8Explain how the parts of a cell work together to enhance cellular function.

Concept 6.1: Biologists use microscopes and biochemistry to study cells

How can cell biologists investigate the inner workings of a cell, usually too small to be seen by the unaided eye? Before we tour the cell, it will be helpful to learn how cells are studied.

Microscopy

The development of instruments that extend the human senses allowed the discovery and early study of cells. Microscopes were invented in 1590 and further refined during the 1600s. Cell walls were first seen on dead cells of oak bark by Robert Hooke in 1665 and living cells by Antoni van Leeuwenhoek a few years later.

The microscopes first used by Renaissance scientists, as well as the microscopes you are likely to use in the laboratory, are all light microscopes. In a **light microscope(LM)**, visible light is passed through the specimen and then through glass lenses. The lenses refract (bend) the light in such a way that the image of the specimen is magnified as it is projected into the eye or into a camera (see Appendix C).

Three important parameters in microscopy are magnification, resolution, and contrast. *Magnification* is the ratio of an object's image size to its real size. Light microscopes can magnify effectively to about 1,000 times the actual size of the specimen; at greater magnifications, additional details cannot be seen clearly. *Resolution* is a measure of the clarity of the image; it is the minimum distance two points can be separated and still be distinguished as separate points. For example, what appears to the unaided eye as one star in the sky may be resolved as twin stars with a telescope, which has a higher resolving ability than the eye. Similarly, using standard techniques, the light microscope cannot resolve detail finer than about 0.2 micrometer (μm), or 200 nanometers (nm), regardless of the magnification **(Figure 6.2)**. The third parameter, *contrast*, is the difference in brightness between the light and dark areas of an image. Methods for enhancing contrast include staining or labeling cell components to stand out visually. **Figure 6.3** shows some different types of microscopy; study this figure as you read this section.

Figure 6.2

The size range of cells.

A chart illustrates size range of cells with scale from just below the atom (0.1 nanometers) to above human height (10 meters), with reference lines marking tenfold difference in length or diameter.

The chart reads as follows. Scale key:

- 1 centimeter = 10 to the minus 2 of a meter = 0.4 inches

- 1 millimeter = 10 to the minus 3 of a meter
- 1 micrometer = 10 to the minus 3 millimeter or 10 to the minus 6 meter
- 1 nanometer = 10 to the minus 3 of a micrometer or 10 to the minus 9 of a meter.

Visible by the unaided human eye:Human height indicated at about 6 meters; length of some muscle and nerve cells (about 0.8 meters); Chicken egg (about 5 centimeters); Frog egg (about 2 millimeters). On the scale of visibility with both the light microscope and the electron microscope are the human egg cell (about 100 micrometers); most plant and animal cells (range from 10 to 100 micrometers); the cell nucleus (about 8 micrometers), cell mitochondria (about 2 micrometers) and most bacteria (range from 1 to 10 micrometers). The smallest bacteria (just over 100 nanometers), viruses (70 to 100 nanometers) and ribosomes (12 nanometers) can be seen with super-resolution light microscopes or an electron microscope. Visible with an electron microscope only are large molecules like proteins (5 to 10 nanometers); and lipids (2 to 5 nanometers). Too small to be seen even with electron microscopes are small molecules (0.5 to 1 nanometer) and atoms (0.2 nanometers).

Most cells are between 1 and 100 μm in diameter (yellow region of chart), and their components are even smaller (see Figure 6.32), as are viruses. Notice that the scale along the left side is logarithmic, to accommodate the range of sizes shown. Starting at the top of the scale with 10 m and going down, each reference measurement marks a tenfold decrease in diameter or length. For a complete table of the metric system, see the back of the book.

Animation

Metric System Review

image

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image

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Figure 6.3

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Exploring Microscopy
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Exploring microscopy.

Types of microscopy are as follows.

- Brightfield (unstained specimen). Light passes directly through the specimen. Unless the cell is naturally pigmented or artificially stained, the image has little contrast. (The first four light micrographs show human cheek epithelial cells; the scale bar pertains to all four micrographs.)
- Brightfield (stained specimen). Staining with various dyes enhances contrast. Most staining procedures require that cells be fixed (preserved), thereby killing them.
- Phase-contrast. Variations in density within the specimen are amplified to enhance contrast in unstained cells; this is especially useful for examining living, unpigmented cells.
- Differential interference contrast (Nomarski). As in phase-contrast microscopy, optical modifications are used to exaggerate differences in density; the image appears almost 3-D.
- Fluorescence. The locations of specific molecules in the cell can be revealed by labeling the molecules with fluorescent dyes or antibodies; some cells have molecules that fluoresce on their own. Fluorescent substances absorb ultraviolet radiation and emit visible light. In this fluorescently labeled uterine cell, nuclear material is blue, organelles called mitochondria are orange, and the cell's skeleton is green.
- Confocal. The top image is a standard fluorescence micrograph of fluorescently labeled nervous tissue (nerve cells are green, support cells are orange, and regions of overlap are yellow); below it is a confocal image of the same tissue. Using a laser, this optical sectioning technique eliminates out-of-focus light from a thick sample, creating a single plane of fluorescence in the image. By capturing sharp images at many different planes, a 3-D reconstruction can be created. The standard image is blurry because out-of-focus light is not excluded.
- Deconvolution. The top of this split image is a compilation of standard fluorescence micrographs through the depth of a white blood cell. Below is an image of the same cell reconstructed from many blurry images at different planes, each of which was processed using deconvolution software. This process digitally removes out-of-focus light and reassigns it to its source, creating a much sharper 3-D image.
- Super-resolution. On the top is a confocal image of part of a nerve cell, using a fluorescent label that binds to a molecule clustered in small sacs in the cell (vesicles) that are 40 nm in diameter. The greenish-yellow spots are blurry because 40 nm is below the 200-nm limit of resolution for standard light microscopy. Below is an image of the same part of the cell, seen using a new super-resolution technique. Sophisticated equipment is used to light up individual fluorescent molecules and record their position. Combining information from many molecules in different places breaks the limit of resolution, resulting in the sharp greenish-yellow dots seen here. (Each dot is a_40-nm vesicle.)
- Scanning electron microscopy (SEM). Micrographs taken with a scanning electron microscope show a 3-D image of the surface of a specimen. This SEM shows the surface of a cell from a trachea (windpipe) covered with cell projections called cilia. Electron micrographs are black and white but are often artificially colorized to highlight particular structures, as has been done with both electron micrographs shown here.

- Transmission electron microscopy (TEM). A transmission electron microscope profiles a thin section of a specimen. This TEM shows a section through a tracheal cell, revealing its internal structure. In preparing the specimen, some cilia were cut along their lengths, creating longitudinal sections, while other cilia were cut straight across, creating cross sections. Abbreviations used in figure legends in this text: LM = Light Micrograph. SEM = Scanning Electron Micrograph. TEM = Transmission Electron Micrograph

VISUAL SKILLS *When the tissue was sliced, what was the orientation of the cilia in the lower portion of the TEM? The upper portion? Explain how the orientation of the cilia determined the type of sections we see.*

*For suggested answer, see*Appendix A.**

Until recently, the resolution barrier prevented cell biologists from using standard light microscopy when studying **organelles**, the membrane-enclosed structures within eukaryotic cells. To see these structures in any detail required the development of a new instrument. In the 1950s, the electron microscope was introduced to biology. Rather than focusing light, the **electron microscope(EM)** focuses a beam of electrons through the specimen or onto its surface (see Appendix C). Resolution is inversely related to the wavelength of the light (or electrons) a microscope uses for imaging, and electron beams have much shorter wavelengths than visible light. Modern electron microscopes can theoretically achieve a resolution of about 0.002 nm, though in practice they usually cannot resolve structures smaller than about 2 nm across. Still, this is a 100-fold improvement over the standard light microscope.

The **scanning electron microscope(SEM)** is especially useful for detailed study of the topography of a specimen (see Figure 6.3). The electron beam scans the surface of the sample, usually coated with a thin film of gold. The beam excites electrons on the surface, and these secondary electrons are detected by a device that translates the pattern of electrons into an electronic signal sent to a video screen. The result is an image of the specimen's surface that appears three-dimensional.

The **transmission electron microscope(TEM)** is used to study the internal structure of cells (see Figure 6.3). The TEM aims an electron beam through a very thin section of the specimen, much as a light microscope aims light through a sample on a slide. For the TEM, the specimen has been stained with atoms of heavy metals, which attach to certain cellular structures, thus enhancing the electron density of some parts of the cell more than others. The electrons passing through the specimen are scattered more in the denser regions, so fewer are transmitted. The image displays the pattern of transmitted electrons. Instead of using glass lenses, both the SEM and TEM use electromagnets as lenses to bend the paths of the electrons, ultimately focusing the image onto a monitor for viewing.

Electron microscopes have revealed many subcellular structures that were impossible to resolve with the light microscope. But the light microscope offers advantages, especially in studying living cells. A

disadvantage of electron microscopy is that the methods customarily used to prepare the specimen kill the cells and can introduce artifacts, structural features seen in micrographs that do not exist in the living cell.

In the past several decades, light microscopy has been revitalized by major technical advances (see Figure 6.3). Labeling individual cellular molecules or structures with fluorescent markers has made it possible to see such structures with increasing detail. In addition, both confocal and deconvolution microscopy have produced sharper images of three-dimensional tissues and cells. Finally, a group of new techniques and labeling molecules developed in recent years, called *super-resolution microscopy*, has allowed researchers to "break" the resolution barrier and distinguish subcellular structures as small as 10-20 nm across.

A recently developed new type of TEM called cryo-electron microscopy (cryo-EM) (see Figure 6.3) allows specimens to be preserved at extremely low temperatures. This avoids the use of preservatives, allowing visualization of structures in their cellular environment. This method is increasingly used to complement X-ray crystallography in revealing protein complexes and subcellular structures like ribosomes, described later. Cryo-EM has even been used to resolve some individual proteins. The Nobel Prize for Chemistry was awarded in 2017 to the developers of this valuable technique.

Microscopes are the most important tools of *cytology*, the study of cell structure. Understanding the function of each structure, however, required the integration of cytology and *biochemistry*, the study of the chemical processes (metabolism) of cells.

Cell Fractionation

A useful technique for studying cell structure and function is **cell fractionation(Figure 6.4)**, which takes cells apart and separates major organelles and other subcellular structures from one another. The piece of equipment that is used for this task is the centrifuge, which spins test tubes holding mixtures of disrupted cells at a series of increasing speeds, a process called *differential centrifugation*. At each speed, the resulting force causes a subset of the cell components to settle to the bottom of the tube, forming a pellet. At lower speeds, the pellet consists of larger components, and higher speeds result in a pellet with smaller components.

Figure 6.4

Research Method: Cell Fractionation

Application

Cell fractionation is used to separate (fractionate) cell components based on size and density.

Technique

Cells are homogenized in a blender to break them up. The resulting mixture (*homogenate*) is centrifuged. The liquid above the pellet (*supernatant*) is poured into another tube and centrifuged at a higher speed for a longer period. This process is repeated several times. This process, called *differential centrifugation*, results in a series of pellets, each containing different cell components.

A diagram of cell fractionation.

Tissue cells are homogenized in a blender to create homogenate, which is placed in a centrifuge. It is centrifuged at 1,000 times the force of gravity for 10 minutes, resulting in a pellet in the tube rich in nuclei and cellular debris. The supernatant from the first tube is poured into the next tube, which is centrifuged at 2,000 times the force of gravity for 20 minutes, resulting in a pellet rich in mitochondria and chloroplasts if cells are from a plant. The second tube is poured into a third which is centrifuged at 80,000 times the force of gravity for 60 minutes, resulting in a pellet rich in microsomes, pieces of plasma membranes and cells internal membranes. The third tube is poured into a fourth, which is centrifuged at 150,000 times the force of gravity for 3 hours, resulting in a pellet rich in ribosomes.

Results

In early experiments, researchers used microscopy to identify the organelles in each pellet and biochemical methods to determine their metabolic functions. These identifications established a baseline for this method, enabling today's researchers to know which cell fraction they should collect in order to isolate and study particular organelles.

MAKE CONNECTIONS*If you wanted to study the process of translation of proteins from mRNA, which part of which fraction would you use? (See*Figure 5.22.)*

*For suggested answer, see*Appendix A.*

Cell fractionation enables researchers to prepare specific cell components in bulk and identify their functions, a task not usually possible with intact cells. For example, on one of the cell fractions, biochemical tests showed the presence of enzymes involved in cellular respiration, while electron microscopy revealed large numbers of the organelles called mitochondria. Together, these data helped biologists determine that mitochondria are the sites of cellular respiration. Biochemistry and cytology thus complement each other in correlating cell function with structure.

Concept Check 6.1

- How do stains used for light microscopy compare with those used for electron microscopy?

- **WHAT IF?** Which type of microscope would you use to study (a) the changes in shape of a living white blood cell? (b) the details of surface texture of a hair?

*For suggested answers, see*Appendix A.**

Concept 6.2: Eukaryotic cells have internal membranes that compartmentalize their functions

Cells—the basic structural and functional units of every organism—are of two distinct types: prokaryotic and eukaryotic. Organisms of the domains Bacteria and Archaea consist of prokaryotic cells. Organisms of the domain Eukarya—protists, fungi, animals, and plants—all consist of eukaryotic cells. ("Protist" is an informal term referring to a diverse group of mostly unicellular eukaryotes.)

Comparing Prokaryotic and Eukaryotic Cells

All cells share certain basic features: They are all bounded by a selective barrier, called the *plasma membrane* (or the cell membrane). Inside all cells is a semifluid, jellylike substance called **cytosol**, in which subcellular components are suspended. All cells contain *chromosomes*, which carry genes in the form of DNA. And all cells have *ribosomes*, tiny complexes that make proteins according to instructions from the genes.

A major difference between prokaryotic and eukaryotic cells is the location of their DNA. In a **eukaryotic cell**, most of the DNA is in an organelle called the *nucleus*, which is bounded by a double membrane (see Figure 6.8). In a **prokaryotic cell**, the DNA is concentrated in a region that is not membrane-enclosed, called the **nucleoid(Figure 6.5)**. *Eukaryotic* means "true nucleus" (from the Greek *eu*, true, and *karyon*, kernel, referring to the nucleus), and *prokaryotic* means "before nucleus" (from the Greek *pro*, before), reflecting the earlier evolution of prokaryotic cells.

Figure 6.5

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A prokaryotic cell.
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A prokaryotic cell.

Lacking a true nucleus and the other membrane-enclosed organelles of the eukaryotic cell, the prokaryotic cell appears much simpler in internal structure. Prokaryotes include bacteria and archaea; the general cell

structure of these two domains is quite similar.

Figure a is a diagram of a prokaryotic cell, a typical rod-shaped bacterium, and it is compared to figure b, a TEM of a thin section through the bacterium Bacillus coagulans (scale 0.5 micrometers). Structures are labeled on the bacterial diagram and TEM as follows:

- Fimbriae: (shown on the diagram only) hair-like attachment structures on the surface of some prokaryotes, here cover the surface of the cell.
- Nucleoid: region where the cell's DNA is located (not enclosed by a membrane). In the diagram, a large region composed of a tangled string-like structure that is the bacterial chromosome. In the TEM, a dark region in the middle of the cell is visible.
- Ribosomes: complexes that synthesize proteins. In both the diagram and the TEM seen as small dots.
- Plasma membrane: a membrane enclosing the cytoplasm, just inside the cell wall.
- Cell wall: a rigid structure outside the plasma membrane.
- Capsule: jellylike outer coating of many prokaryotes; the outer portion of both the diagram and the TEM.
- Flagella: long, whip-like structures extending from one end of the bacterial cell in the diagram. Not shown in the TEM. Locomotion structures of some bacteria.

Animation

Prokaryotic Cell Structure and Function

image

image

The interior of either type of cell is called the **cytoplasm**; in eukaryotic cells, this term refers only to the region between the nucleus and the plasma membrane. Within the cytoplasm of a eukaryotic cell, suspended in cytosol, are a variety of organelles of specialized form and function. These membrane-bounded structures are absent in almost all prokaryotic cells, another distinction between prokaryotic and eukaryotic cells. In spite of the absence of organelles, though, the prokaryotic cytoplasm is not a formless soup. For example, some prokaryotes contain regions surrounded by proteins (not membranes), within which specific reactions take place.

Eukaryotic cells are generally much larger than prokaryotic cells (see Figure 6.2). Size is a general feature of cell structure that relates to function. The logistics of carrying out cellular metabolism sets limits on cell size. At the lower limit, the smallest cells known are bacteria called mycoplasmas, which have diameters between 0.1 and 1.0μm. These are perhaps the smallest packages with enough DNA to program metabolism and enough enzymes and other cellular equipment to carry out the activities necessary for a cell to sustain itself

and reproduce. Typical bacteria are 1–5μm in diameter, about ten times the size of mycoplasmas. Eukaryotic cells are typically 10–100μm in diameter.

Metabolic requirements also impose theoretical upper limits on the size that is practical for a single cell. At the boundary of every cell, the **plasma membrane** functions as a selective barrier that allows passage of enough oxygen, nutrients, and wastes to service the entire cell **(Figure 6.6)**. For each square micrometer of membrane, only a limited amount of a particular substance can cross per second, so the ratio of surface area to volume is critical. As a cell (or any other object) increases in size, its surface area grows proportionately less than its volume. (Area is proportional to a linear dimension squared, whereas volume is proportional to the linear dimension cubed.) Thus, a smaller cell has a greater ratio of surface area to volume: Compare the calculations for the first two "cells" in **Figure 6.7**. The **Scientific Skills Exercise** gives you a chance to calculate the volumes and surface areas of two actual cells—a mature yeast cell and a cell budding from it. To explore different ways that the surface area of cells is maximized in various organisms, see Make Connections Figure 33.8.

Figure 6.6

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The plasma membrane.
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The plasma membrane.

The plasma membrane and the membranes of organelles consist of a double layer (bilayer) of phospholipids with various proteins attached to or embedded in it. The hydrophobic parts of phospholipids and membrane proteins are found in the interior of the membrane, while the hydrophilic parts are in contact with aqueous solutions on either side. Carbohydrate side chains may be attached to proteins or lipids on the outer surface of the plasma membrane.

Figure a. TEM of a plasma membrane. The plasma membrane, here in a red blood cell, appears as a pair of dark bands separated by a light band. (Scale: 0.1 micrometers).

Figure b. Structure of the plasma membrane, consisting of hydrophilic and hydrophobic regions. Hydrophilic regions exist on both sides of the membrane, consisting of a mat of spherical polar heads of phospholipids. The hydrophobic region is the center of the bilayer, with the lipid tails of the phospholipids appearing as parallel lines tucked into the interior of the membrane. Membrane proteins are embedded in the membrane spanning both layers, while carbohydrate side chains stick out from the surface of the cell.

VISUAL SKILLS *What parts of the membrane diagram in (b) correspond to the dark bands in the TEM in (a)? What parts correspond to the gold band? (Review*Figure 5.11.)**

*For suggested answer, see*Appendix A.**

BioFlix® Animation

Membranes

image

Play

image

Figure 6.7

Geometric relationships between surface area and volume.

Geometric relationships between surface area and volume.

A cube with sides of 5 units has a surface area that increases while the total volume remains constant as it is divided into smaller cubes with sides of 1 unit. Each face of the larger cube is 5 by 5.

A table is as follows:

Total surface area [sum of the surface areas (height times width) of all box sides times number of boxes]	6	150
Total volume [height times width times length times number of boxes]	1	125
Surface-to-volume (S-to-V) ratio [surface area divided by volume]	6	12

In this diagram, cells are represented as cubes. Using arbitrary units of length, we can calculate the cell's surface area (in square units, or units2), volume (in cubic units, or units3), and ratio of surface area to volume. A high surface area-to-volume ratio facilitates the exchange of materials between a cell and its environment.

Figure 6.7 Walkthrough

Geometric relationships between surface area and volume

Watch an explanation of this figure in the Figure 6.7 Walkthrough: Geometric relationships between surface area and volume.

image

image

**

Scientific Skills Exercise

Using a Scale Bar to Calculate Volume and Surface Area of a Cell

AP Science Practices 2, 5

How Much New Cytoplasm and Plasma Membrane Are Made by a Growing Yeast Cell? The unicellular yeast *Saccharomyces cerevisiae* divides by budding off a small new cell that then grows to full size (see the yeast cells at the bottom of Figure 6.8). During its growth, the new cell synthesizes new cytoplasm, which increases its volume, and new plasma membrane, which increases its surface area. In this exercise, you will use a scale bar to determine the sizes of a mature parent yeast cell and a cell budding from it. You will then calculate the volume and surface area of each cell. You will use your calculations to determine how much cytoplasm and plasma membrane the new cell needs to synthesize to grow to full size.

How the Experiment Was Done Yeast cells were grown under conditions that promoted division by budding. The cells were then viewed with a differential interference contrast light microscope and photographed.

Data from the Experiment This light micrograph shows a budding yeast cell about to be released from the mature parent cell:

A cell buds from a mature parent cell; the budding cell is smaller and appears fused with the parent cell on one side. Scale is 1 micrometer.

Micrograph from K. Tatchell, using yeast cells grown for experiments described in L. Kozubowski et al., Role of the septin ring in the asymmetric localization of proteins at the mother-bud neck in *Saccharomyces cerevisiae, Molecular Biology of the Cell* 16:3455–3466 (2005).

Micrograph from K. Tatchell, using yeast cells grown for experiments described in L. Kozubowski et al. Role of the septin ring in the asymmetric localization of proteins at the mother-bud neck in Saccharomyces cerevisiae, Molecule Biology of the Cell 16:3455-3466 (2005).

INTERPRET THE DATA

- Examine the micrograph of the yeast cells. The scale bar under the photo is labeled 1μm. The scale bar works in the same way as a scale on a map, where, for example, 1 inch equals 1 mile. In this case the bar represents one thousandth of a millimeter. Using the scale bar as a basic unit, determine the diameter of the mature parent cell and the new cell. Start by measuring the scale bar and the

diameter of each cell. The units you use are irrelevant, but working in millimeters is convenient. Divide each diameter by the length of the scale bar and then multiply by the scale bar's length value to give you the diameter in micrometers.

- The shape of a yeast cell can be approximated by a sphere.
- ◦ Calculate the volume of each cell using the formula for the volume of a sphere:

V=43πr3

A sphere's diameter is a segment from one boundary to another, passing through the center. The radius is the distance from one boundary to the center.

- ◦ Note that π (the Greek letter pi) is a constant with an approximate value of 3.14, d stands for diameter, and r stands for radius, which is half the diameter.
- ◦ What volume of new cytoplasm will the new cell have to synthesize as it matures? To determine this, calculate the difference between the volume of the full-sized cell and the volume of the new cell.
- As the new cell grows, its plasma membrane needs to expand to contain the increased volume of the cell.
- Calculate the surface area of each cell using the formula for the surface area of a sphere: A=4πr2.
- How much area of new plasma membrane will the new cell have to synthesize as it matures?
- When the new cell matures, it will be approximately how many times greater in volume and how many times greater in surface area than its current size?

Instructors: A version of this Scientific Skills Exercise can be assigned in **Mastering Biology**.

The need for a surface area large enough to accommodate the volume helps explain the microscopic size of most cells and the narrow, elongated shapes of some cells, such as nerve cells. Larger organisms do not generally have *larger* cells than smaller organisms—they simply have *more* cells (see the far right of Figure 6.7). A sufficiently high ratio of surface area to volume is especially important in cells that exchange a lot of material with their surroundings, such as intestinal cells. Such cells may have many long, thin projections from their surface called *microvilli*, which increase surface area without an appreciable increase in volume.

The evolutionary relationships between prokaryotic and eukaryotic cells will be discussed later in this chapter, and prokaryotic cells will be described in detail elsewhere (see Chapter 27). Most of the discussion of cell structure that follows in this chapter applies to eukaryotic cells.

A Panoramic View of the Eukaryotic Cell

In addition to the plasma membrane at its outer surface, a eukaryotic cell has extensive, elaborately arranged internal membranes that divide the cell into compartments—the organelles mentioned earlier. The cell's compartments provide different local environments that support specific metabolic functions, so incompatible processes can occur simultaneously in a single cell. The plasma membrane and organelle membranes also participate directly in the cell's metabolism because many enzymes are built right into the membranes.

The basic fabric of most biological membranes is a double layer of phospholipids and other lipids. Embedded in this lipid bilayer or attached to its surfaces are diverse proteins (see Figure 6.6). However, each type of membrane has a unique composition of lipids and proteins suited to that membrane's specific functions. For example, enzymes embedded in the membranes of the organelles called mitochondria function in cellular respiration. Because membranes are so fundamental to the organization of the cell, Chapter 7 will discuss them in detail.

Before continuing with this chapter, examine the eukaryotic cells in **Figure 6.8**. The generalized diagrams of an animal cell and a plant cell introduce the various organelles and show the key differences between animal and plant cells. The micrographs at the bottom of the figure give you a glimpse of cells from different types of eukaryotic organisms.

Figure 6.8

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Exploring Eukaryotic Cells
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A cutaway of a generalized animal cell.

Parts of the cell are labeled as follows.

- Nucleus. Consists of the nuclear envelope enclosing the nucleolus and chromatin. The nucleus comprises nearly 1/3 of the cell in this diagram.
- Nuclear envelope: double membrane enclosing the nucleus; perforated by pores; continuous with E R.
- Nucleolus: nonmembranous structure involved in production of ribosomes; a nucleus has one or more nucleoli.
- Chromatin: material consisting of D N A and proteins; visible in a dividing cell as individual condensed chromosomes.
- Endoplasmic reticulum (E R): network of membranous sacs and tubes; active in membrane synthesis and other synthetic and metabolic processes; has rough (ribosome-studded) and smooth

regions. (Rough E R) Continuous with the nuclear envelope, the rough E R appears as many folded layers studded with dot-like ribosomes. (Smooth E R) Beyond the rough E R, the smooth E R is much smaller (about 1/5 the size of the rough E R) and has a thick web-like appearance in this diagram.

- Golgi apparatus: organelle active in synthesis, modification, sorting, and secretion of cell products. (Golgi apparatus) Between the rough E R and the plasma membrane, the Golgi apparatus is also drawn as a heavily folded structure similar in size to the smooth E R.
- Mitochondrion: organelle where cellular respiration occurs and most A T P is generated. (Mitochondrion) Kidney bean shaped structures, about half the size of the Golgi apparatus, drawn in cross section so that the inner folds (cristae) are visible. Several shown.
- Lysosome: digestive organelle where macromolecules are hydrolyzed. Small spherical structure about one-fourth the size of the mitochondria.
- Peroxisome: organelle with various specialized metabolic functions; produces hydrogen peroxide as a by-product and then converts it to water. Small spherical structure similar in size to the lysosome.
- Ribosomes (small brown dots): complexes that make proteins; free in cytosol or bound to rough E R or nuclear envelope.
- Cytoskeleton: reinforces cell's shape; functions in cell movement; components are made of protein. Includes microfilaments, intermediate filaments, and microtubules. (Cytoskeleton) Drawn as long tubules or thick filaments or string-like structures and composed in order of decreasing thickness: the microtubules, the intermediate filaments, and the microfilaments.
- Centrosome: region where the cell's microtubules are initiated; contains a pair of centrioles.
- Plasma membrane: membrane enclosing the cell.
- Flagellum. Appears as a whip-like structure protruding from the cell; length is similar to cell length as drawn.
- Flagellum: motility structure present in some animal cells, composed of a cluster of microtubules within an extension of the plasma membrane.
- Microvilli: projections that increase the cell's surface area.
- Animal cells. Human cells from the lining of the uterus (colorized TEM) with nucleus about 10 micrometers in diameter and nucleolus visible as a dark spot within the nucleus.

A cutaway view of a generalized plant cell.

Parts of the plant cell are labeled as follows.

- Chloroplast: photosynthetic organelle; converts energy of sunlight to chemical energy stored in sugar molecules. Composed of the microfilaments and the thicker microtubules.
- Plasmodesmata: cytoplasmic channels through cell walls that connect the cytoplasms of adjacent cell.

- Cell wall: outer layer that maintains cell's shape and protects cell from mechanical damage; made of cellulose, other polysaccharides, and protein.
- Central vacuole. Drawn here in the center of the cell, a large structure comparable in size to the nucleus. Central vacuole: prominent organelle in older plant cells; functions include storage, breakdown of waste products, and hydrolysis of macromolecules; enlargement of the vacuole is a major mechanism of plant growth.
- Golgi apparatus. Drawn here between the central vacuole and the plasma membrane, the Golgi apparatus is a heavily folded structure about one-third the size of the central vacuole.
- Nucleus. The nuclear envelope encloses the nucleolus and chromatin.
- Plant cells. Cells from duckweed (Spirodela oligorrhiza), a floating plant (colorized TEM). Cell walls separate cells containing distinct nuclei with nucleoli, chloroplasts and mitochondria. Scale: 5 micrometers.

Two micrographs are as follows.

- Unicellular Eukaryotes. Colorized SEM: Unicellular green alga Chlamydomonas (cell diameter = about 8 micrometers) with two flagella visible.
- Unicellular Eukaryotes. Colorized TEM (scale: 1 micrometer): In slice view labeled are the nucleus with dark nucleolus, six chloroplasts, a vacuole, cell wall, and two flagella.

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Tour of an Animal Cell

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Tour of a Plant Cell

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Video

Turgid *Elodea*

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Chlamydomonas

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Concept Check 6.2

- Briefly describe the structure and function of the nucleus, the mitochondrion, the chloroplast, and the endoplasmic reticulum.
- **DRAW IT** Draw a simplified elongated cell that measures 125×1×1 arbitrary units. A nerve cell would be roughly this shape. Predict how its surface-to-volume ratio would compare with those in Figure 6.7. Then calculate the ratio and check your prediction.

*For suggested answers, see*Appendix A.**

Concept 6.3: The eukaryotic cell's genetic instructions are housed in the nucleus and carried out by the ribosomes

On the first stop of our detailed tour of the eukaryotic cell, let's look at two cellular components involved in the genetic control of the cell: the nucleus, which houses most of the cell's DNA, and the ribosomes, which use information from the DNA to make proteins.

The nucleus is colorized on a 5 micrometer scale.

The Nucleus: Information Central

The **nucleus** contains most of the genes in the eukaryotic cell. (Some genes are located in mitochondria and chloroplasts.) It is usually the most conspicuous organelle (see the purple structure in the fluorescence micrograph), averaging about 5μm in diameter. The **nuclear envelope** encloses the nucleus **(Figure 6.9)**, separating its contents from the cytoplasm.

Figure 6.9

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The nucleus and its envelope.
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The nucleus and its envelopes.

Within the nucleus are the chromosomes, which appear as a mass of chromatin (DNA and associated proteins) and one or more nucleoli (singular, nucleolus), which function in ribosome synthesis. The nuclear envelope, which consists of two membranes separated by a narrow space, is perforated with pores and lined by the nuclear lamina.

Drawing of a cross-section of an animal cell with nucleus highlighted. Arrows indicate zooming in on structures in and associated with the nucleus, with TEMs illustrating some structures. Parts are as follows.

Arrow from cell cross section points to large drawing of the nucleus, including nuclear envelope studded with pore complexes, and the inner and outer membranes labeled; chromatin drawn as strands within the nucleus, and the nucleolus a large, dark region inside the nucleus. The rough E R is adjacent to the nucleus.

Arrows from the cell nucleus also zoom in on the surface of the nuclear envelope in a TEM, prepared by a technique known as freeze-fracture. Inner and outer membranes as well as nuclear pores (raised studs) are labeled.

Arrows from the pore complexes indicate a drawing and TEM zoomed in on these structures. Pore complexes drawn as openings in the nuclear envelope with ribosomes nearby. On a TEM pore complexes appear as star-like structures, each pore ringed by protein particles (scale: 0.25 micrometers).

Arrows from the pore complex drawing zoom in on a TEM of the nuclear lamina (scale: 0.5 micrometers). The netlike lamina lines the inner surface of the nuclear envelope.

Arrow from the nucleus zooms in on chromatin. This segment of a chromosome from non-dividing cells shows two states of coiling of the D N A (string-like) and protein (thick) complex. The thicker form is sometimes organized into long loops.

MAKE CONNECTIONS *Chromosomes contain the genetic material and reside in the nucleus. How does the rest of the cell get access to the information they carry? (See*Figure 5.22.)**

*For suggested answer, see*Appendix A.**

The nuclear envelope is a *double* membrane. The two membranes, each a lipid bilayer with associated proteins, are separated by a space of 20–40 nm. The envelope is perforated by pore structures that are about 100 nm in diameter. At the lip of each pore, the inner and outer membranes of the nuclear envelope are continuous. An intricate protein structure called a *pore complex* lines each pore and plays an important role in the cell by regulating the entry and exit of proteins and RNAs, as well as large complexes of macromolecules. Except at the pores, the nuclear side of the envelope is lined by the **nuclear lamina**, a netlike array of protein filaments (in animal cells, called *intermediate filaments*) that maintains the shape of the nucleus by mechanically supporting the nuclear envelope. There is also much evidence for a *nuclear matrix*, a framework of protein fibers extending throughout the nuclear interior. The nuclear lamina and matrix may help organize the genetic material so it functions efficiently.

Within the nucleus, the DNA is organized into discrete units called **chromosomes**, structures that carry the genetic information. Each chromosome contains one long DNA molecule associated with many proteins, including small basic proteins called histones. Some of the proteins help coil the DNA molecule of each chromosome, reducing its length and allowing it to fit into the nucleus. The complex of DNA and proteins making up chromosomes is called **chromatin**. When a cell is not dividing, stained chromatin appears as a diffuse mass in micrographs, and the chromosomes cannot be distinguished from one another, even though discrete chromosomes are present. As a cell prepares to divide, however, the chromosomes form loops and coil, condensing and becoming thick enough to be distinguished under a microscope as separate structures (see Figure 16.23). Each eukaryotic species has a characteristic number of chromosomes. For example, a typical human cell has 46 chromosomes in its nucleus; the exceptions are human sex cells (eggs and sperm), which have only 23 chromosomes. A fruit fly cell has 8 chromosomes in most cells and 4 in the sex cells.

A prominent structure within the nondividing nucleus is the **nucleolus** (plural, *nucleoli*), which appears through the electron microscope as a mass of densely stained granules and fibers adjoining part of the chromatin. Here a type of RNA called *ribosomal RNA* (rRNA) is synthesized from genes in the DNA. Also in the nucleolus, proteins imported from the cytoplasm are assembled with rRNA into large and small subunits of ribosomes. These subunits then exit the nucleus through the nuclear pores to the cytoplasm, where a large and a small subunit can assemble into a ribosome. Sometimes there are two or more nucleoli; the number depends on the species and the stage in the cell's reproductive cycle. The nucleoli may also play a role in controlling cell division and the life span of a cell.

As we saw in Figure 5.22, the nucleus directs protein synthesis by synthesizing messenger RNA (mRNA) that carries information from the DNA. The mRNA is then transported to the cytoplasm via nuclear pores. Once an mRNA molecule reaches the cytoplasm, ribosomes translate the mRNA's genetic message into the primary structure of a specific polypeptide. (This process of transcribing and translating genetic information is described in detail in Chapter 17.)

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Nucleus and Ribosomes

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Ribosomes: Protein Factories

Ribosomes, which are complexes made of ribosomal RNAs and proteins, are the cellular components that carry out protein synthesis **(Figure 6.10)**. (Note that ribosomes are not membrane bounded and thus are not considered organelles.) Cells with high rates of protein synthesis have particularly large numbers of ribosomes as well as prominent nucleoli, which makes sense, given the role of nucleoli in ribosome assembly. For example, a human pancreas cell, which makes many digestive enzymes, has a few million ribosomes.

Figure 6.10

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Ribosomes.
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Diagram of animal cell with Rough ER and free ribosomes highlighted. Arrows zoom in on TEM of these structures, and flow to a diagram and computer model of the ribosome.

This electron micrograph of a pancreas cell shows both free and bound ribosomes. The simplified diagram and computer model show the two subunits of a ribosome.

Diagram of cell shows the Rough E R surrounding the nucleus highlighted, as well as ribosomes free in the cytosol. Both bound and free ribosomes are visible in a TEM (scale: 0.25 micrometers); ribosomes appear as dark dots; E R appears as bands in the image.

Arrows flow from ribosomes in the TEM to a diagram of a ribosome, which consists of two globular shapes fitted together: a large subunit nestled on top of a smaller subunit. Both subunits of the ribosome are also distinguishable in a computer model of a ribosome, with each subunit made up of masses of ribbon-like structures.

DRAW IT *After you have read the section on ribosomes, circle a ribosome in the micrograph that might be making a protein that will be secreted.*

*For suggested answer, see*Appendix A.**

Interview

Interview with Venki Ramakrishnan: Studying ribosome structure

Venki Ramakrishnan.

Ribosomes build proteins in two cytoplasmic regions: At any given time, *free ribosomes* are suspended in the cytosol, while *bound ribosomes* are attached to the outside of the endoplasmic reticulum or nuclear envelope (see Figure 6.10). Bound and free ribosomes are structurally identical, and ribosomes can play either role at different times. Most of the proteins made on free ribosomes function within the cytosol; examples are enzymes that catalyze the first steps of sugar breakdown. Bound ribosomes generally make proteins that are destined for insertion into membranes, for packaging within certain organelles such as lysosomes (see Figure 6.8), or for export from the cell (secretion). Cells that specialize in protein secretion—for instance, the cells of the pancreas that secrete digestive enzymes—frequently have a high proportion of bound ribosomes. (You will learn more about ribosome structure and function in Concept 17.4.)

Concept Check 6.3

- What role do ribosomes play in carrying out genetic instructions?
- Describe the molecular composition of nucleoli and explain their function.
- **WHAT IF?** As a cell begins the process of dividing, its chromosomes become shorter, thicker, and individually visible in an LM (light micrograph). Explain what is happening at the molecular level.

*For suggested answers, see*Appendix A.**

Concept 6.4: The endomembrane system regulates protein traffic and performs metabolic functions

Many of the different membrane-bounded organelles of the eukaryotic cell are part of the **endomembrane system**, which includes the nuclear envelope, the endoplasmic reticulum, the Golgi apparatus, lysosomes, various kinds of vesicles and vacuoles, and the plasma membrane. This system carries out a variety of tasks in the cell, including synthesis of proteins, transport of proteins into membranes and organelles or out of the cell, metabolism and movement of lipids, and detoxification of poisons. The membranes of this system are related either through direct physical continuity or by the transfer of membrane segments as tiny **vesicles** (sacs made of membrane). Despite these relationships, the various membranes are not identical in structure

and function. Moreover, the thickness, molecular composition, and types of chemical reactions carried out in a given membrane are not fixed, but may be modified several times during the membrane's life. Having already looked at the nuclear envelope, we will now focus on the endoplasmic reticulum and the other endomembranes to which the endoplasmic reticulum gives rise.

The Endoplasmic Reticulum: Biosynthetic Factory

The **endoplasmic reticulum(ER)** is such an extensive network of membranes that it accounts for more than half the total membrane in many eukaryotic cells. (The word *endoplasmic* means "within the cytoplasm," and *reticulum* is Latin for "little net.") The ER consists of a network of membranous tubules and sacs called cisternae (from the Latin *cisterna*, a reservoir for a liquid). The ER membrane separates the internal compartment of the ER, called the *ER lumen* (cavity) or cisternal space, from the cytosol. And because the ER membrane is continuous with the nuclear envelope, the space between the two membranes of the envelope is continuous with the lumen of the ER **(Figure 6.11)**.

Figure 6.11

Endoplasmic reticulum (ER).

TEM image: Smooth E R appears heavily folded or convoluted while Rough E R appears as regular bands dotted with ribosomes. Scale = 0.20 micrometers.

A membranous system of interconnected tubules and flattened sacs called cisternae, the ER is also continuous with the nuclear envelope, as shown in the cutaway diagram at the top. The membrane of the ER encloses a continuous compartment called the ER lumen (or cisternal space). Rough ER, which is studded on its outer surface with ribosomes, can be distinguished from smooth ER in the electron micrograph (TEM). Transport vesicles bud off from a region of the rough ER called transitional ER and travel to the Golgi apparatus and other destinations.

There are two distinct, though connected, regions of the ER that differ in structure and function: smooth ER and rough ER. **Smooth ER** is so named because its outer surface lacks ribosomes. **Rough ER** is studded with ribosomes on the outer surface of the membrane and thus appears rough through the electron microscope. As already mentioned, ribosomes are also attached to the cytoplasmic side of the nuclear envelope's outer membrane, which is continuous with rough ER.

Functions of Smooth ER

The smooth ER functions in diverse metabolic processes, which vary with cell type. These processes include synthesis of lipids, metabolism of carbohydrates, detoxification of drugs and poisons, and storage of calcium ions.

Enzymes of the smooth ER are important in the synthesis of lipids, including oils, steroids, and new membrane phospholipids. Among the steroids produced by the smooth ER in animal cells are the sex hormones of vertebrates and the various steroid hormones secreted by the adrenal glands. The cells that synthesize and secrete these hormones—in the testes and ovaries, for example—are rich in smooth ER, a structural feature that fits the function of these cells.

Other enzymes of the smooth ER help detoxify drugs and poisons, especially in liver cells. Detoxification usually involves adding hydroxyl groups to drug molecules, making them more water-soluble and easier to flush from the body. The sedative phenobarbital and other barbiturates are examples of drugs metabolized in this manner by smooth ER in liver cells. In fact, barbiturates, alcohol, and many other drugs induce the proliferation of smooth ER and its associated detoxification enzymes, thus increasing the rate of detoxification. This, in turn, increases tolerance to the drugs, meaning that higher doses are required to achieve a particular effect, such as sedation. Also, because some of the detoxification enzymes have relatively broad action, the proliferation of smooth ER in response to one drug can increase the need for higher dosages of other drugs as well. Barbiturate use, for instance, can decrease the effectiveness of certain antibiotics and other useful drugs.

The smooth ER also stores calcium ions. In muscle cells, for example, the smooth ER membrane pumps calcium ions from the cytosol into the ER lumen. When a muscle cell is stimulated by a nerve impulse, calcium ions rush back across the ER membrane into the cytosol and trigger contraction of the muscle cell. In other cell types, release of calcium ions from the smooth ER triggers different responses, such as secretion of vesicles carrying newly synthesized proteins.

Functions of Rough ER

Many cells secrete proteins that are produced by ribosomes attached to rough ER. For instance, certain pancreatic cells synthesize the protein insulin in the ER and secrete this hormone into the bloodstream. As a polypeptide chain grows from a bound ribosome, the chain is threaded into the ER lumen through a pore formed by a protein complex in the ER membrane. The new polypeptide folds into its functional shape as it enters the ER lumen. Most secretory proteins are **glycoproteins**, proteins with carbohydrates covalently bonded to them. The carbohydrates are attached to the proteins in the ER lumen by enzymes built into the ER membrane.

After secretory proteins are formed, the ER membrane keeps them separate from proteins in the cytosol, which are produced by free ribosomes. Secretory proteins depart from the ER wrapped in the membranes of vesicles that bud like bubbles from a specialized region called transitional ER (see Figure 6.11). Vesicles in transit from one part of the cell to another are called **transport vesicles**; we will examine their fate shortly.

In addition to making secretory proteins, rough ER is a membrane factory for the cell; it grows in place by adding membrane proteins and phospholipids to its own membrane. As polypeptides destined to be

membrane proteins grow from the ribosomes, they are inserted into the ER membrane itself and anchored there by their hydrophobic portions. Like the smooth ER, the rough ER also makes membrane phospholipids; enzymes built into the ER membrane assemble phospholipids from precursors in the cytosol. The ER membrane expands, and portions of it are transferred in the form of transport vesicles to other components of the endomembrane system.

The Golgi Apparatus: Shipping and Receiving Center

After leaving the ER, many transport vesicles travel to the **Golgi apparatus**. We can think of the Golgi as a warehouse for receiving, sorting, shipping, and even some manufacturing. Here, products of the ER, such as proteins, are modified and stored and then sent to other destinations. Not surprisingly, the Golgi apparatus is especially extensive in cells specialized for secretion.

The Golgi apparatus consists of a group of associated, flattened membranous sacs—cisternae—looking like a stack of pita bread **(Figure 6.12)**. A cell may have many, even hundreds, of these stacks. The membrane of each cisterna in a stack separates its internal space from the cytosol. Vesicles concentrated in the vicinity of the Golgi apparatus are engaged in the transfer of material between parts of the Golgi and other structures.

Figure 6.12

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The Golgi apparatus.
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Cutaway diagram of Golgi apparatus with corresponding TEM (scale 0.1 micrometers). Vesicles move from the cis face (receiving side) to the trans face (shipping side) of the Golgi apparatus.

The Golgi apparatus consists of stacks of associated, flattened sacs, or cisternae. Unlike the ER cisternae, these sacs are not physically connected, as you can see in the cutaway diagram. A Golgi stack receives and dispatches transport vesicles and the products they contain. A Golgi stack has a structural and functional directionality, with a cis face that receives vesicles containing ER products and a trans face that dispatches vesicles. The cisternal maturation model proposes that the Golgi cisternae themselves "mature," moving from the cis to the trans face while carrying some proteins along. In addition, some vesicles recycle enzymes that had been carried forward in moving cisternae, transporting them "backward" to a less mature region where their functions are needed.

Labeled parts are as follows.

- Vesicles (with contents) move from the E R to the Golgi.
- Vesicles coalesce (fuse with Golgi) to form new cis Golgi cisternae.
- Cisternal maturation: Golgi cisternae move in a cis to trans direction.

- Vesicles form and leave Golgi, carrying specific products to other locations or to the plasma membrane for secretion.
- Vesicles transport some proteins backward to less mature Golgi cisternae, where they function.
- Vesicles also transport certain proteins back to the E R, their site of function.

A Golgi stack has a distinct structural directionality, with the membranes of cisternae on opposite sides of the stack differing in thickness and molecular composition. The two sides of a Golgi stack are referred to as the *cis* face and the *trans* face; these act, respectively, as the receiving and shipping departments of the Golgi apparatus. The term *cis* means "on the same side," and the *cis* face is usually located near the ER. Transport vesicles move material from the ER to the Golgi apparatus. A vesicle that buds from the ER can add its membrane and the contents of its lumen to the *cis* face by fusing with a Golgi membrane on that side. The *trans* face ("on the opposite side") gives rise to vesicles that pinch off and travel to other sites.

Products of the endoplasmic reticulum are usually modified during their transit from the *cis* region to the *trans* region of the Golgi apparatus. For example, glycoproteins formed in the ER have their carbohydrates modified, first in the ER itself, and then as they pass through the Golgi. The Golgi removes some sugar monomers and substitutes others, producing a large variety of carbohydrates. Membrane phospholipids may also be altered in the Golgi.

In addition to its finishing work, the Golgi apparatus also manufactures some macromolecules. Many polysaccharides secreted by cells are Golgi products. For example, pectins and certain other noncellulose polysaccharides are made in the Golgi of plant cells and then incorporated along with cellulose into their cell walls. Like secretory proteins, nonprotein Golgi products that will be secreted depart from the *trans* face of the Golgi inside transport vesicles that eventually fuse with the plasma membrane. The contents are released and the vesicle membrane is incorporated into the plasma membrane, adding to the surface area.

The Golgi manufactures and refines its products in stages, with different cisternae containing unique teams of enzymes. Until recently, biologists viewed the Golgi as a static structure, with products in various stages of processing transferred from one cisterna to the next by vesicles. While this may occur, research from several labs has given rise to a new model of the Golgi as a more dynamic structure. According to the *cisternal maturation model*, the cisternae of the Golgi actually progress forward from the *cis* to the *trans* face, carrying and modifying their cargo as they move. Figure 6.12 shows the details of this model. The reality probably lies somewhere between the two models; recent research suggests the central regions of the cisternae may remain in place, while the outer ends are more dynamic.

Before a Golgi stack dispatches its products by budding vesicles from the *trans* face, it sorts these products and targets them for various parts of the cell. Molecular identification tags, such as phosphate groups added to the Golgi products, aid in sorting by acting like zip codes on mailing labels. Finally, transport vesicles budded from the Golgi may have external molecules on their membranes that recognize "docking sites" on the surface of specific organelles or on the plasma membrane, thus targeting the vesicles appropriately.

Lysosomes: Digestive Compartments

A **lysosome** is a membranous sac of hydrolytic enzymes that many eukaryotic cells use to digest (hydrolyze) macromolecules. Lysosomal enzymes work best in the acidic environment found in lysosomes. If a lysosome breaks open or leaks its contents, the released enzymes are not very active because the cytosol has a near-neutral pH. However, excessive leakage from a large number of lysosomes can destroy a cell by self-digestion.

Hydrolytic enzymes and lysosomal membrane are made by rough ER and then transferred to the Golgi apparatus for further processing. At least some lysosomes probably arise by budding from the *trans* face of the Golgi apparatus (see Figure 6.12). How are the proteins of the inner surface of the lysosomal membrane and the digestive enzymes themselves spared from destruction? Apparently, the three-dimensional shapes of these proteins protect vulnerable bonds from enzymatic attack.

Lysosomes carry out intracellular digestion in a variety of circumstances. Amoebas and many other unicellular protists eat by engulfing smaller organisms or food particles, a process called **phagocytosis** (from the Greek *phagein*, to eat, and *kytos*, vessel, referring here to the cell). The *food vacuole* formed in this way then fuses with a lysosome, whose enzymes digest the food (**Figure 6.13a**, top). Digestion products, including simple sugars, amino acids, and other monomers, pass into the cytosol and become nutrients for the cell. Some human cells also carry out phagocytosis. Among them are macrophages, a type of white blood cell that helps defend the body by engulfing and destroying bacteria and other invaders (see Figure 6.13a, bottom, and Figure 6.31).

Figure 6.13

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Lysosomes.
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Lysosomes and the processes of phagocytosis and autophagy.

Part a. Phagocytosis. In phagocytosis, lysosomes digest (hydrolyze) materials taken into the cell. Top: In this macrophage (a type of white blood cell) from a rat, the lysosomes are very dark because of a stain that reacts with one of the products of digestion inside the lysosome (TEM). Macrophages ingest bacteria and viruses and destroy them using lysosomes. Bottom: This diagram shows a lysosome fusing with a food vacuole during the process of phagocytosis by a unicellular eukaryote. Shown as a three-step process:

- lysosome contains active hydrolytic enzymes.
- lysosome fuses with food vacuole.
- hydrolytic enzymes digest food particles.

Part b. The cytoplasm of this rat liver cell is a vesicle containing two disabled organelles (TEM). The vesicle will fuse with a lysosome in the process of autophagy, which recycles intracellular materials. Bottom: This diagram shows fusion of such a vesicle with a lysosome. This type of vesicle has a double membrane of unknown origin. The outer membrane fuses with the lysosome, and the inner membrane is degraded along with the damaged organelles. Shown as two steps:

- lysosome fuses with vesicle containing damaged organelles.
- hydrolytic enzymes digest organelle components.

Lysosomes also use their hydrolytic enzymes to recycle the cell's own organic material, a process called *autophagy*. During autophagy, a damaged organelle or small amount of cytosol becomes surrounded by a double membrane (of unknown origin), and a lysosome fuses with the outer membrane of this vesicle **(Figure 6.13b)**. The lysosomal enzymes dismantle the inner membrane and the enclosed material, and the resulting small organic compounds are released to the cytosol for reuse. With the help of lysosomes, the cell continually renews itself. A human liver cell, for example, recycles half of its macromolecules each week.

The cells of people with inherited lysosomal storage diseases lack a functioning hydrolytic enzyme normally present in lysosomes. The lysosomes become engorged with indigestible material, which begins to interfere with other cellular activities. In Tay-Sachs disease, for example, a lipid-digesting enzyme is missing or inactive, and the brain becomes impaired by an accumulation of lipids in the cells. Fortunately, lysosomal storage diseases are rare in the general population.

Vacuoles: Diverse Maintenance Compartments

Vacuoles are large vesicles derived from the endoplasmic reticulum and Golgi apparatus. Thus, vacuoles are an integral part of a cell's endomembrane system. Like all cellular membranes, the vacuolar membrane is selective in transporting solutes; as a result, the solution inside a vacuole differs in composition from the cytosol.

Vacuoles perform a variety of functions in different kinds of cells. **Food vacuoles**, formed by phagocytosis, have already been mentioned (see Figure 6.13a). Many unicellular protists living in fresh water have **contractile vacuoles** that pump excess water out of the cell, thereby maintaining a suitable concentration of ions and molecules inside the cell (see Figure 7.13). In plants and fungi, certain vacuoles carry out enzymatic hydrolysis, a function shared by lysosomes in animal cells. (In fact, some biologists consider these hydrolytic vacuoles to be a type of lysosome.) In plants, small vacuoles can hold reserves of important organic compounds, such as the proteins stockpiled in the storage cells in seeds. Vacuoles may also help protect the plant against herbivores by storing compounds that are poisonous or unpalatable to animals. Some plant vacuoles contain pigments, such as the red and blue pigments of petals that help attract pollinating insects to flowers.

Mature plant cells generally contain a large **central vacuole(Figure 6.14)**, which develops by the coalescence of smaller vacuoles. The solution inside the central vacuole, called cell sap, is the plant cell's main repository of inorganic ions, including potassium and chloride. The central vacuole plays a major role in the growth of plant cells, which enlarge as the vacuole absorbs water, enabling the cell to become larger with a minimal investment in new cytoplasm. The cytosol often occupies only a thin layer between the central vacuole and the plasma membrane, so the ratio of plasma membrane surface to cytosolic volume is sufficient, even for a large plant cell.

Figure 6.14

The plant cell vacuole.

Drawing of plant cell with arrows zooming in on TEM. Central vacuole, cell wall, nucleus and cytosol all labeled. Central vacuole has a diameter of about 5 micrometers.

The central vacuole is usually the largest compartment in a plant cell; the rest of the cytoplasm is often confined to a narrow zone between the vacuolar membrane and the plasma membrane (TEM).

BioFlix® Animation

Central Vacuole

image

Play

image

Play

The Endomembrane System: *A Review*

Figure 6.15 reviews the endomembrane system, showing the flow of membrane lipids and proteins through the various organelles. As the membrane moves from the ER to the Golgi and then elsewhere, its molecular composition and metabolic functions are modified, along with those of its contents. The endomembrane system is a complex and dynamic player in the cell's compartmental organization.

Figure 6.15

```
    Review: Relationships among organelles of the endomembrane
system.
```

Relationships shown among nuclear envelope, endoplasmic reticulum, Golgi apparatus, lysosomes, transport vesicles, and plasma membrane.

The red arrows show some of the migration pathways for membranes and the materials they enclose.

Labeled parts are as follows.

- The nuclear envelope is connected to the rough E R, which is also continuous with the smooth E R.
- Membranes and proteins produced by the ER move via transport vesicles to the Golgi.
- The Golgi pinches off transport vesicles and other vesicles that give rise to lysosomes, other types of specialized vesicles, and vacuoles.
- The lysosome is available for fusion with another vesicle for digestion.
- A transport vesicle carries proteins to the plasma membrane for secretion.
- The plasma membrane expands by fusion of vesicles; proteins are secreted from the cell.

BioFlix® Animation

Endomembrane System

image

Play

image

Play

We'll continue our tour of the cell with some organelles that are not closely related to the endomembrane system but play crucial roles in the energy transformations carried out by cells.

Concept Check 6.4

- Describe the structural and functional distinctions between rough and smooth ER.
- Describe how transport vesicles integrate the endomembrane system.
- **WHAT IF?** Imagine a protein that functions in the ER but requires modification in the Golgi apparatus before it can achieve that function. Describe the protein's path through the cell, starting with the mRNA molecule that specifies the protein.

*For suggested answers, see*Appendix A.**

Concept 6.5: Mitochondria and chloroplasts change energy from one form to another

Organisms transform the energy they acquire from their surroundings. In eukaryotic cells, mitochondria and chloroplasts are the organelles that convert energy to forms that cells can use for work. ********Mitochondria** (singular, *mitochondrion*) are the sites of cellular respiration, the metabolic process that uses oxygen to drive the generation of ATP by extracting energy from sugars, fats, and other fuels. **Chloroplasts**, found in plants and algae, are the sites of photosynthesis. This process in chloroplasts converts solar energy to chemical energy by absorbing sunlight and using it to drive the synthesis of organic compounds such as sugars from carbon dioxide and water.

In addition to having related functions, mitochondria and chloroplasts share similar evolutionary origins, which we'll look at briefly before examining their structures. In this section, we'll also consider the peroxisome, an oxidative organelle. The evolutionary origin of the peroxisome, as well as its relation to other organelles, is still a matter of some debate.

The Evolutionary Origins of Mitochondria and Chloroplasts

EVOLUTION Mitochondria and chloroplasts display similarities with bacteria that led to the **endosymbiont theory**, illustrated in **Figure 6.16**. This theory states that an early ancestor of eukaryotic cells (a *host* cell) engulfed an oxygen-using nonphotosynthetic prokaryotic cell. Eventually, the engulfed cell formed a relationship with the host cell in which it was enclosed, becoming an *endosymbiont* (a cell living within another cell). Indeed, over the course of evolution, the host cell and its endosymbiont merged into a single organism, a eukaryotic cell with the endosymbiont having become a mitochondrion. At least one of these cells may have then taken up a photosynthetic prokaryote, becoming the ancestor of eukaryotic cells that contain chloroplasts.

Figure 6.16

```
    The endosymbiont theory of the origins of mitochondria and
chloroplasts in eukaryotic cells.
```

The endosymbiotic theory of the origin of mitochondria and chloroplasts in eukaryotic cells is diagrammed.

According to this theory, the proposed ancestors of mitochondria were oxygen-using nonphotosynthetic prokaryotes that were taken into host cells, while the proposed ancestors of chloroplasts were photosynthetic prokaryotes. The large arrows represent change over evolutionary time; the small arrows inside the cells show the process of the endosymbiont becoming an organelle, also over long periods of time.

Ancestor of eukaryotic cells (host cell) drawn containing the nucleus with the nuclear envelope continuous with the Endoplasmic reticulum. Large arrow points to the host cell, which contains the nucleus and E R. Small arrows within the cell indicate the process of engulfing a formerly free-living cell oxygen-using prokaryote which, over many generations of cells, becomes a mitochondrion. This cell is now labeled a nonphotosynthetic eukaryote. Large arrow points to a cell (at least one) labeled a photosynthetic eukaryote, which contains the nucleus, E R, and mitochondrion. Small arrows within the cell indicate the process of the host cell engulfing a formerly free-living photosynthetic prokaryote, which over time becomes a chloroplast. This cell is now labeled a photosynthetic eukaryote.

This is a widely accepted theory, which we will examine in more detail in Concept 25.3. This theory is consistent with many structural features of mitochondria and chloroplasts. First, rather than being bounded by a single membrane like organelles of the endomembrane system, mitochondria and typical chloroplasts have two membranes surrounding them. (Chloroplasts also have an internal system of membranous sacs.) There is evidence that the ancestral engulfed prokaryotes had two outer membranes, which became the double membranes of mitochondria and chloroplasts. Second, like prokaryotes, mitochondria and chloroplasts contain ribosomes, as well as circular DNA molecules—like bacterial chromosomes—associated with their inner membranes. The DNA in these organelles programs the synthesis of some organelle proteins on ribosomes that have been synthesized and assembled there as well. Third, also consistent with their probable evolutionary origins as cells, mitochondria and chloroplasts are autonomous (somewhat independent) organelles that grow and reproduce within the cell.

Next, we focus on the structures of mitochondria and chloroplasts, while providing an overview of their structures and functions. (In Chapters 9 and 10, we will examine their roles as energy transformers.)

Mitochondria: Chemical Energy Conversion

Mitochondria are found in nearly all eukaryotic cells, including those of plants, animals, fungi, and most protists. Some cells have a single large mitochondrion, but more often a cell has hundreds or even thousands of mitochondria; the number correlates with the cell's level of metabolic activity. For example, cells that move or contract have proportionally more mitochondria per volume than less active cells.

Each of the two membranes enclosing the mitochondrion is a phospholipid bilayer with a unique collection of embedded proteins. The outer membrane is smooth, but the inner membrane is convoluted, with infoldings called **cristae(Figure 6.17a)**. The inner membrane divides the mitochondrion into two internal

compartments. The first is the intermembrane space, the narrow region between the inner and outer membranes. The second compartment, the **mitochondrial matrix**, is enclosed by the inner membrane. The matrix contains many different enzymes as well as the mitochondrial DNA and ribosomes. Enzymes in the matrix catalyze some of the steps of cellular respiration. Other proteins that function in respiration, including the enzyme that makes ATP, are built into the inner membrane. As highly folded surfaces, the cristae give the inner mitochondrial membrane a large surface area, thus enhancing the productivity of cellular respiration. This is another example of structure fitting function. (Chapter 9 discusses cellular respiration in detail.)

Figure 6.17

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The mitochondrion, site of cellular respiration.
```

Two diagrams of mitochondria.

(a) *The inner and outer membranes seen in the drawing and TEM establish two compartments: the intermembrane space and the mitochondrial matrix. The cristae increase the surface area of the inner membrane. Circular DNA molecules are associated with the inner mitochondrial membrane.* ***(b)*** *Mitochondria are dynamic, at times separating into fragments or fusing.* ***(c)*** *The LM shows one protist (Euglena gracilis) at a much lower magnification than the TEM. The mitochondria form a branched tubular network; the mitochondrial matrix is stained green. The nuclear DNA is stained red; molecules of mitochondrial DNA appear as bright yellow spots.*

Part a. Cristae of mitochondria visible as regular bands in the TEM image. Each of the cristae appear just under 0.1 micrometers apart.

Part b. Points of mitochondrial DNA scattered throughout Euglena cell. Mitochondria appear as tubules connected throughout the cell. Scale bar shows 10 micrometers; the length of the longest mitochondria.

BioFlix® Animation

Mitochondria

image

Play

image

Play

Mitochondria are generally in the range of 1–10 μm long. Time-lapse films of living cells reveal mitochondria moving around, changing their shapes, and fusing or dividing into separate fragments, unlike the static structures seen in most diagrams and electron micrographs. These studies helped cell biologists understand that mitochondria in a living cell form a branched tubular network that is in a dynamic state of flux (see **Figure 6.17b** and **c**). In skeletal muscle, this network has been referred to by researchers as a "power grid."

Chloroplasts: Capture of Light Energy

Chloroplasts contain the green pigment chlorophyll, along with enzymes and other molecules that function in the photosynthetic production of sugar. These lens-shaped organelles, about 3–6 μm. in length, are found in leaves and other green organs of plants and in algae (**Figure 6.18**; see also Figure 6.26c).

Figure 6.18

```
The chloroplast, site of photosynthesis.
```

Two diagrams are given.

(a) *Many plants have lens-shaped chloroplasts, as shown here in a diagram and a TEM. A chloroplast has three compartments: the intermembrane space, the stroma, and the thylakoid space. Free ribosomes are present in the stroma, as are copies of chloroplast DNA molecules.* ***(b)*** *This light micrograph, at a much lower magnification than the TEM, shows a whole cell of the green alga Spirogyra crassa, which is named for the spiral arrangement of each cell's chloroplasts.*

Part a. Grana, inner and outer membranes, and stroma visible on diagram and TEM image. Ribosomes are also drawn into the diagram. Scale shows chloroplast about 5 micrometers long.

Part b. A micrograph shows chloroplasts scattered throughout an algal cell; scale 50 micrometers.

BioFlix® Animation

Chloroplasts and Mitochondria

image

Play

image

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The contents of a chloroplast are partitioned from the cytosol by an envelope consisting of two membranes separated by a very narrow intermembrane space. Inside the chloroplast is another membranous system in the form of flattened, interconnected sacs called **thylakoids**. In some regions, thylakoids are stacked like poker chips; each stack is called a **granum** (plural, *grana*). The fluid outside the thylakoids is the **stroma**, which contains the chloroplast DNA and ribosomes as well as many enzymes. The membranes of the chloroplast divide the chloroplast space into three compartments: the intermembrane space, the stroma, and the thylakoid space. This compartmental organization enables the chloroplast to convert light energy to chemical energy during photosynthesis. (You will learn more about photosynthesis in Chapter 10.)

As with mitochondria, the static and rigid appearance of chloroplasts in micrographs or schematic diagrams cannot accurately depict their dynamic behavior in the living cell. Their shape is changeable, and they grow and occasionally pinch in two, reproducing themselves. They are mobile and, with mitochondria and other organelles, move around the cell along tracks of the cytoskeleton, a structural network we will consider in Concept 6.6.

The chloroplast is a specialized member of a family of closely related plant organelles called **plastids**. One type of plastid, the *amyloplast*, is a colorless organelle that stores starch (amylose), particularly in roots and tubers. Another is the *chromoplast*, which has pigments that give fruits and flowers their orange and yellow hues.

Peroxisomes: Oxidation

The **peroxisome** is a specialized metabolic compartment bounded by a single membrane **(Figure 6.19)**. Peroxisomes contain enzymes that remove hydrogen atoms from various substrates and transfer them to oxygen O2, producing hydrogen peroxide H2O2 as a by-product (from which the organelle derives its name). These reactions have many different functions. Some peroxisomes use oxygen to break fatty acids down into smaller molecules that are transported to mitochondria and used as fuel for cellular respiration. Peroxisomes in the liver detoxify alcohol and other harmful compounds by transferring hydrogen from the poisonous compounds to oxygen. The H2O2 formed by peroxisomes is itself toxic, but the organelle also contains an enzyme that converts H2O2 to water. This is an excellent example of how the cell's compartmental structure is crucial to its functions: The enzymes that produce H2O2 and those that dispose of this toxic compound are sequestered away from other cellular components that could be damaged.

Figure 6.19

```
A peroxisome.
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TEM of peroxisome with square crystalline core visible, between several chloroplasts (grana inside) and a mitochondrion. Mitochondrion is about 1 micrometer in diameter; the peroxisome is about four times larger.

Peroxisomes are roughly spherical and often have a granular or crystalline core that is thought to be a dense collection of enzyme molecules. Chloroplasts and mitochondria cooperate with peroxisomes in certain metabolic functions (TEM).

Specialized peroxisomes called *glyoxysomes* are found in the fat-storing tissues of plant seeds. These organelles contain enzymes that initiate the conversion of fatty acids to sugar, which the emerging seedling uses as a source of energy and carbon until it can produce its own sugar by photosynthesis.

Peroxisomes grow larger by incorporating proteins made in the cytosol and ER, as well as lipids made in the ER and within the peroxisome itself. But how peroxisomes increase in number and how they arose in evolution—as well as what organelles they are related to—are still open questions.

Concept Check 6.5

- Describe two characteristics shared by chloroplasts and mitochondria. Consider both function and membrane structure.
- Do plant cells have mitochondria? Explain.
- **WHAT IF?** A classmate proposes that mitochondria and chloroplasts should be classified in the endomembrane system. Argue against the proposal.

*For suggested answers, see*Appendix A.**

Concept 6.6: The cytoskeleton is a network of fibers that organizes structures and activities in the cell

In the early days of electron microscopy, biologists thought that the organelles of a eukaryotic cell floated freely in the cytosol. But improvements in both light microscopy and electron microscopy have revealed the **cytoskeleton**, a network of fibers extending throughout the cytoplasm **(Figure 6.20)**. Bacterial cells also have fibers that form a type of cytoskeleton, constructed of proteins similar to eukaryotic ones, but here we will concentrate on eukaryotes. The eukaryotic cytoskeleton plays a major role in organizing the structures and activities of the cell.

Figure 6.20

```
The cytoskeleton.
```

Microtubules appear as a network of string-like structures concentrated around the nucleus (diameter: about 8 micrometers) and microfilaments are clustered toward the perimeter of the cell.

As shown in this fluorescence micrograph, the cytoskeleton extends throughout the cell. The cytoskeletal elements have been tagged with different fluorescent molecules: green for microtubules and reddish-orange for microfilaments. A third component of the cytoskeleton, intermediate filaments, is not evident here. (The blue color tags the DNA in the nucleus.)

BioFlix® Animation

Cytoskeleton

image

Play

image

Roles of the Cytoskeleton: Support and Motility

The most obvious function of the cytoskeleton is to give mechanical support to the cell and maintain its shape. This is especially important for animal cells, which lack walls. The remarkable strength and resilience of the cytoskeleton as a whole are based on its architecture. Like a dome tent, the cytoskeleton is stabilized by a balance between opposing forces exerted by its elements. And just as the skeleton of an animal helps fix the positions of other body parts, the cytoskeleton provides anchorage for many organelles and even cytosolic enzyme molecules. The cytoskeleton is more dynamic than an animal skeleton, however. It can be quickly dismantled in one part of the cell and reassembled in a new location, changing the shape of the cell.

Some types of cell motility (movement) also involve the cytoskeleton. The term *cell motility* includes both changes in cell location and movements of cell parts. Cell motility generally requires interaction of the cytoskeleton with **motor proteins**. There are many such examples: Cytoskeletal elements and motor proteins work together with plasma membrane molecules to allow whole cells to move along fibers outside the cell. Inside the cell, vesicles and other organelles often use motor protein “feet” to “walk” to their destinations along a track provided by the cytoskeleton. For example, this is how vesicles containing neurotransmitter molecules migrate to the tips of axons, the long extensions of nerve cells that release these

molecules as chemical signals to adjacent nerve cells **(Figure 6.21)**. The cytoskeleton also manipulates the plasma membrane, bending it inward to form food vacuoles or other phagocytic vesicles.

Figure 6.21

Motor proteins and the cytoskeleton.

Motor proteins and the cytoskeleton.

Part a. Vesicle attached to motor protein at a receptor point. Arrows from an A T P indicate that a motor protein steps along microtubule using energy from A T P. Microtubule drawn as hollow tube.

Part b. Two vesicles containing neurotransmitters move along microtubule toward tip of a nerve cell extension called an axon (SEM). Microtubule visible as cable-like structure. Vesicles are globular structures atop the microtubule. Vesicles are globular structures approximately 0.25 micrometers long atop the microtubule.

Interview

Interview with George Langford: Investigating how motor proteins move cargo

George Langford.

Components of the Cytoskeleton

Now let's look more closely at the three main types of fibers that make up the cytoskeleton: *Microtubules* are the thickest of the three types; *microfilaments* (also called actin filaments) are the thinnest; and *intermediate filaments* are fibers with diameters in a middle range **(Table 6.1)**.

Table 6.1

```
The Structure and Function of the Cytoskeleton
```

A table with three fluorescent micrographs of fibroblasts are given.

Property	Microtubules (Tubulin polymers)	Microfilaments (actin filaments)	Intermediate filaments
Structure	Hollow tubes	Two intertwined strands of actin	Fibrous proteins coiled into cables

Diameter	25 nanometers with 15 nanometer lumen	7 nanometers	8 to 12 nanometers
Protein subunits	Tubulin, a dimer consisting of an alpha tubulin and a beta tubulin	Actin	One of several different proteins (including keratins)
Main functions	Maintenance of cell shape; cell motility; chromosome movements in cell division; organelle movements	Maintenance of cell shape; changes in cell shape; muscle contraction; cytoplasmic streaming (plant cells); cell motility; cell division (animal cells)	Maintenance of cell shape; anchorage of nucleus and certain other organelles; formation of nuclear lamina
Fluorescence micrographs of fibroblasts. Connective tissue cells called fibroblasts are a favorite cell type for cell biology studies because they spread out flat and their internal structures are easy to see. In each fibroblast shown here, the structure of interest has been tagged with fluorescent molecules. The D N A in the nucleus has also been tagged in the first micrograph (blue) and third	Image shows cluster of 7 fibroblasts (diameter of each cell: about 10 micrometers) filled with networks of microtubules. Below image is a diagram of a hollow tube representing a microtubule and a cutaway drawing zooming in on microtubule structure as a coil of columns of tubulin dimers. Tubulin dimers are made of two subunits: alpha and beta tubulin. Diameter of the microtubule = 25 nanometers.	Actin filaments networked across a fibroblast (diameter about 10 micrometers). Diagram zooms in on filament shown as a coil of two thick strands. Zooming in on the structure reveals a coil of spherical actin subunits pushed together like beads (diameter 7 nanometers).	Image of fibroblast (about 10 micrometers in diameter) with intermediate filaments and nucleus visible (about 5 micrometers in diameter). Cutaway diagrams zoom in on intermediate filament structure. Intermediate filament structure shown as a thick tube made up of keratin proteins with fibrous subunits (keratins coiled together). Filament diameter = 8-12 nanometers.

micrograph (orange).			

VISUAL SKILLS *How many tubulin dimers are in the boxed row?*

*For suggested answer, see*Appendix A.**

Microtubules

All eukaryotic cells have **microtubules**, hollow rods constructed from globular proteins called tubulins. Each tubulin protein is a *dimer*, a molecule made up of two components. A tubulin dimer consists of two slightly different polypeptides, α-tubulin and β-tubulin. Microtubules grow in length by adding tubulin dimers; they can also be disassembled and their tubulins used to build microtubules elsewhere in the cell. Because of the orientation of tubulin dimers, the two ends of a microtubule are slightly different. One end can accumulate or release tubulin dimers at a much higher rate than the other, thus growing and shrinking significantly during cellular activities. (This is called the "plus end," not because it can only add tubulin proteins but because it's the end where both "on" and "off" rates are much higher.)

Microtubules shape and support the cell and also serve as tracks along which organelles equipped with motor proteins can move. In addition to the example in Figure 6.21, microtubules guide vesicles from the ER to the Golgi apparatus and from the Golgi to the plasma membrane. Microtubules are also involved in the separation of chromosomes during cell division, as shown in Figure 12.7.

Centrosomes and Centrioles

In animal cells, microtubules grow out from a **centrosome**, a region that is often located near the nucleus. These microtubules function as compression-resisting girders of the cytoskeleton. Within the centrosome is a pair of **centrioles**, each composed of nine sets of triplet microtubules arranged in a ring **(Figure 6.22)**. Although centrosomes with centrioles may help organize microtubule assembly in animal cells, many other eukaryotic cells lack centrosomes with centrioles and instead organize microtubules by other means.

Figure 6.22

Centrosome containing a pair of centrioles.

Cutaway diagram of two centrioles at right angles to each other. Structure is a star-shaped tube made up of 9 sets of triplet sets of microtubules separated by non-tubulin proteins.

Included beneath the diagram is an image of two centrioles positioned at right angles to each other in a cell, so that the (columnar) longitudinal section of one centriole is visible, and a star-shaped cross section of the other appears (diameter = about 0.25 micrometers). Networks of single microtubules are visible in the background in the cell.

Most animal cells have a centrosome, a region near the nucleus where the cell's microtubules are initiated. Within the centrosome is a pair of centrioles, each about 250 nm (0.25 μm) in diameter. The two centrioles are at right angles to each other, and each is made up of nine sets of three microtubules. The blue portions of the drawing represent nontubulin proteins that connect the microtubule triplets.

VISUAL SKILLS *How many microtubules are in a centrosome? In the drawing, circle and label one microtubule and describe its structure. Circle and label a triplet.*

*For suggested answer, see*Appendix A.**

Cilia and Flagella

Some eukaryotic cells have **flagella** (singular, *flagellum*) and **cilia** (singular, *cilium*), cellular extensions that contain microtubules. A specialized arrangement of the microtubules is responsible for the beating of these structures. (The bacterial flagellum, shown in Figure 6.5, has a completely different structure.) Many unicellular protists are propelled through water by cilia or flagella that act as locomotor appendages, and the sperm of animals, algae, and some plants have flagella. When cilia or flagella extend from cells that are attached tightly together in a sheet that is part of a tissue layer, they can move fluid over the surface of the tissue. For example, the ciliated lining of the trachea (windpipe) sweeps mucus containing trapped debris out of the lungs (see the EMs in Figure 6.3). In a woman's reproductive tract, the cilia lining the oviducts help move an egg toward the uterus.

Motile cilia usually occur in large numbers on the cell surface. Flagella are usually limited to just one or a few per cell, and they are longer than cilia. Flagella and cilia differ in their beating patterns. A flagellum has an undulating motion like the tail of a fish. In contrast, cilia have alternating power and recovery strokes, much like the oars of a racing crew boat **(Figure 6.23)**.

Figure 6.23

A comparison of the beating of flagella and motile cilia.

A comparison of the beating of flagella and motile cilia.

Part a. Motion of flagella. A flagellum usually undulates, its snakelike motion driving a cell in the same direction as the axis of the flagellum. Propulsion of a human sperm cell is an example of flagellate locomotion (LM).

Part b. Motion of cilia. Cilia have a back-and-forth motion. The rapid power stroke moves the cell in a direction perpendicular to the axis of the cilium. Then, during the slower recovery stroke, the cilium bends and sweeps sideways, closer to the cell surface. A dense nap of cilia, beating at a rate of about 40 to 60 strokes a second, covers this Colpidium, a freshwater protist (colorized SEM).

Video

Flagellum Beating with ATP

image

Play

image

Video

Paramecium Cilia

image

Play

image

A cilium may also act as a signal-receiving “antenna” for the cell. Cilia that have this function are generally nonmotile, and there is only one per cell. (In fact, in vertebrate animals, it appears that almost all cells have such a cilium, which is called a *primary cilium*.) Membrane proteins on this kind of cilium transmit molecular signals from the cell’s environment to its interior, triggering signaling pathways that may lead to changes in the cell’s activities. Cilium-based signaling appears to be crucial to brain function and to embryonic development.

Though different in length, number per cell, and beating pattern, motile cilia and flagella share a common structure. Each motile cilium or flagellum has a group of microtubules sheathed in an extension of the plasma membrane **(Figure 6.24a)**. Nine doublets of microtubules are arranged in a ring with two single microtubules in its center **(Figure 6.24b)**. This arrangement, referred to as the “9+2” pattern, is found in nearly all eukaryotic flagella and motile cilia. (Nonmotile primary cilia have a “9+0” pattern, lacking the central pair of microtubules.) The microtubule assembly of a cilium or flagellum is anchored in the cell by a **basal body**, which is structurally very similar to a centriole, with microtubule triplets in a “9+0” pattern **(Figure 6.24c)**. In fact, in many animals (including humans), the basal body of the fertilizing sperm’s flagellum enters the egg and becomes a centriole.

Figure 6.24

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Structure of a flagellum or motile cilium.
```

Structure of a flagellum or motile cilium.

Part a. Microtubule doublets appear as circle of 9 double rings or figure 8s, with two single rings in the center of the circle (TEM). Radius of ring = about 0.1 micrometers. Animal cell with cilia; arrows flow to TEM image of cilium. Microtubules labeled inside the plasma membrane. Basal body labeled at base; arrows zoom in on this structure in c.

Part b. Microtubule doublets appear as circle of 9 double rings or figure eights, with two single rings in the center of the circle (TEM). Radius of ring = about 0.1 micrometers. A cross section through a motile cilium shows the 9 + 2 arrangement of microtubules (TEM). The outer microtubule doublets are held together with the two central microtubules by flexible cross-linking proteins (blue in art), including the radial spokes. The doublets also have attached motor proteins called dyneins (red in art). Structures labeled on both the TEM image and the diagram of the cross section through the motile cilium as follows: Plasma membrane wraps outside the cilium. Outer microtubule doublets form a ring around the two central microtubules and are held to the central microtubules by radial spokes. Doublets also have attached motor proteins which are Y shaped in the diagram.

Part c. Cross-section of a basal body: 9 triplets (three joined rings) are joined by non-tubulin proteins to form a star-shaped ring. Proteins form spokes from each triplet as in a wheel. Basal body: The nine outer doublets of a cilium or flagellum extend into the basal body, where each doublet joins another microtubule to form a ring of nine triplets. Each triplet is connected to the next by non-tubulin proteins (thinner blue lines in diagram). This is a 9 + 0 arrangement: The two central microtubules are not present because they terminate above the basal body (TEM).

DRAW IT *In (a) and (b), circle and label the central pair of microtubules. In (a), show where they terminate, and explain why they aren't seen in the cross section of the basal body in (c).*

*For suggested answer, see*Appendix A.**

Animation

Cilia and Flagella

image

Play

image

Play

How does the microtubule assembly produce the bending movements of flagella and motile cilia? Bending involves large motor proteins called **dyneins** (red in the diagram in Figure 6.24) that are attached along each outer microtubule doublet. A typical dynein protein has two "feet" that "walk" along the microtubule of

the adjacent doublet, using ATP for energy. One foot maintains contact, while the other releases and reattaches one step farther along the microtubule (see Figure 6.21). The outer doublets and two central microtubules are held together by flexible cross-linking proteins (blue in the diagram in Figure 6.24), and the walking movement is coordinated so that it happens on one side of the circle at a time. If the doublets were not held in place, the walking action would make them slide past each other. Instead, the movements of the dynein feet cause the microtubules—and the organelle as a whole—to bend.

Microfilaments (Actin Filaments)

Microfilaments are thin solid rods. They are also called actin filaments because they are built from molecules of **actin**, a globular protein. A microfilament is a twisted double chain of actin subunits (see Table 6.1). Besides occurring as linear filaments, microfilaments can form structural networks when certain proteins bind along the side of such a filament and allow a new filament to extend as a branch. Like microtubules, microfilaments seem to be present in all eukaryotic cells.

In contrast to the compression-resisting role of microtubules, the structural role of microfilaments in the cytoskeleton is to bear tension (pulling forces). A three-dimensional network formed by microfilaments just to the inside of the plasma membrane (*cortical microfilaments*) helps support the cell's shape (see Figure 6.8). This network gives the outer cytoplasmic layer of a cell, called the **cortex**, the semisolid consistency of a gel, in contrast with the more fluid state of the interior cytoplasm. In some kinds of animal cells, such as nutrient-absorbing intestinal cells, bundles of microfilaments make up the core of microvilli, delicate projections that increase the cell's surface area **(Figure 6.25)**.

Figure 6.25

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A structural role of microfilaments.
```

On a 0.25 nanometer scale, microvilli extend from intermediate filaments, reinforced by microfilaments (actin filaments) and with plasma membranes.

The surface area of this nutrient-absorbing intestinal cell is increased by its many microvilli (singular, microvillus), cellular extensions reinforced by bundles of microfilaments. These actin filaments are anchored to a network of intermediate filaments (TEM).

Microfilaments are well known for their role in cell motility. Thousands of actin filaments and thicker filaments made of a protein called **myosin** interact to cause contraction of muscle cells **(Figure 6.26a)**; muscle contraction is described in detail in Concept 50.5. In the unicellular protist *Amoeba* and some of our white blood cells, localized contractions brought about by actin and myosin are involved in the amoeboid (crawling) movement of the cells. The cell crawls along a surface by extending cellular extensions called

pseudopodia (from the Greek *pseudes*, false, and *pod*, foot) and moving toward them **(Figure 6.26b)**. In plant cells, actin-protein interactions contribute to **cytoplasmic streaming**, a circular flow of cytoplasm within cells **(Figure 6.26c)**. This movement, which is especially common in large plant cells, speeds the movement of organelles and the distribution of materials within the cell.

Figure 6.26

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Microfilaments and motility.
```

Microfilaments and motility.

In these three examples, interactions between actin filaments and motor proteins bring about cell movement.

Part a. Myosin motors in muscle cell contraction. The walking of myosin projections (the so-called heads) drives the parallel myosin and actin filaments past each other so that the actin filaments approach each other in the middle (red arrows). This shortens the muscle cell. Muscle contraction involves the shortening of many muscle cells at the same time (TEM).

Part b. Amoeboid movement. Interaction of actin filaments with myosin causes contraction of the cell, pulling the cell's trailing end (at left) forward (to the right) (LM). Amoeboid cell (diameter: about 500 micrometers) with labels: inner cytoplasm, cortex (outer cytoplasm) gel within actin network, and extending pseudopodium, or projections. Arrows indicate direction of movement toward the right.

Part c. Cytoplasmic streaming in plant cells. A layer of cytoplasm cycles around the cell, moving over tracks of actin filaments. Myosin motors attached to some organelles drive the streaming by interacting with the actin (LM). Plant cells (diameter: about 60 micrometers) with cytoplasm and chloroplasts visible. Arrows indicate the direction of streaming within the cells.

BioFlix® Animation

Actin and Myosin in Muscle Contraction

image

image

Video

Amoeba Pseudopodia

image

Play

image

Video

Cytoplasmic Streaming

image

Play

image

Intermediate Filaments

Intermediate filaments are named for their diameter, which is larger than the diameter of microfilaments but smaller than that of microtubules (see Table 6.1). While microtubules and microfilaments are found in all eukaryotic cells, intermediate filaments are only found in the cells of some animals, including vertebrates. Specialized for bearing tension (like microfilaments), intermediate filaments are a diverse class of cytoskeletal elements. Each type is constructed from a particular molecular subunit belonging to a family of proteins whose members include the keratins. Microtubules and microfilaments, in contrast, are consistent in diameter and composition in all eukaryotic cells.

Intermediate filaments are more permanent fixtures of cells than are microfilaments and microtubules, which are often disassembled and reassembled in various parts of a cell. Even after cells die, intermediate filament networks often persist; for example, the outer layer of our skin consists of dead skin cells full of keratin filaments. Intermediate filaments are especially sturdy and play an important role in reinforcing the shape of a cell and fixing the position of certain organelles. For instance, the nucleus typically sits within a cage made of intermediate filaments. Other intermediate filaments make up the nuclear lamina, which lines the interior of the nuclear envelope (see Figure 6.9). In general, the various kinds of intermediate filaments seem to function together as the permanent framework of the entire cell.

Concept Check 6.6

- Describe how cilia and flagella bend.
- **WHAT IF?** Males who have Kartagener's syndrome are unable to reproduce because of immotile sperm, and they tend to experience lung infections. This condition has a genetic basis. Suggest what the underlying genetic defect might be.

*For suggested answers, see*Appendix A.**

Concept 6.7: Extracellular components and connections between cells help coordinate cellular activities

Having crisscrossed the cell to explore its inner components, we complete our tour of the cell by returning to the surface of this microscopic world, where there are additional structures with important functions. The plasma membrane is usually regarded as the boundary of the living cell, but most cells synthesize and secrete materials extracellularly (to the outside of the cell). Although these materials and the structures they form are outside the cell, their study is important to cell biology because they are involved in a great many essential cellular functions.

Cell Walls of Plants

The **cell wall** is an extracellular structure of plant cells **(Figure 6.27)**. This is one of the features that distinguishes plant cells from animal cells. The wall protects the plant cell, maintains its shape, and prevents excessive uptake of water. At the level of the whole plant, the strong walls of specialized cells hold the plant up against the force of gravity. Prokaryotes, some protists, and fungi also have cell walls, as you saw in Figures 6.5 and 6.8. Those organisms will be discussed in Chapters 27, 28, and 31.

Figure 6.27

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Plant cell walls.
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Plant cells have primary walls, secondary walls, and between the cells are middle lamella. The scale is 1 micrometer.

The drawing shows the relationship between primary and secondary cell walls in several mature plant cells. (Organelles aren't shown because many cells with secondary walls, such as the water-conducting cells, lack organelles.) The TEM shows the cell walls where two cells come together. The multilayered partition between plant cells consists of adjoining walls individually secreted by the cells. Adjacent cells are glued together by a very thin layer called the middle lamella.

Plant cell walls are much thicker than the plasma membrane, ranging from 0.1μm to several micrometers. The exact chemical composition of the wall varies from species to species and even from one cell type to another in the same plant, but the basic design of the wall is consistent. Microfibrils made of the polysaccharide cellulose (see Figure 5.6) are synthesized by an enzyme called cellulose synthase and secreted

to the extracellular space, where they become embedded in a matrix of other polysaccharides and proteins. This combination of materials, strong fibers in a “ground substance” (matrix), is the same basic architectural design found in steel-reinforced concrete and in fiberglass.

A young plant cell first secretes a relatively thin and flexible wall called the **primary cell wall** (see the micrograph in Figure 6.27). Between primary walls of adjacent cells is the **middle lamella**, a thin layer rich in sticky polysaccharides called pectins. The middle lamella glues adjacent cells together. (Pectin is used in cooking as a thickening agent in jams and jellies.) When the cell matures and stops growing, it strengthens its wall. Some plant cells do this simply by secreting hardening substances into the primary wall. Other cells add a **secondary cell wall** between the plasma membrane and the primary wall. The secondary wall, often deposited in several laminated layers, has a strong and durable matrix that affords the cell protection and support. Wood, for example, consists mainly of secondary walls.

The Extracellular Matrix (ECM) of Animal Cells

Although animal cells lack walls akin to those of plant cells, they do have an elaborate **extracellular matrix(ECM)**. The main ingredients of the ECM are glycoproteins and other carbohydrate-containing molecules secreted by the cells. (Recall that glycoproteins are proteins with covalently bonded carbohydrates.) The most abundant glycoprotein in the ECM of most animal cells is **collagen**, which forms strong fibers outside the cells (see Figure 5.18). In fact, collagen accounts for about 40% of the total protein in the human body. The collagen fibers are embedded in a network woven out of **proteoglycans** secreted by cells **(Figure 6.28)**. A proteoglycan molecule consists of a small core protein with many carbohydrate chains covalently attached, so that it may be up to 95% carbohydrate. Large proteoglycan complexes can form when hundreds of proteoglycan molecules become noncovalently attached to a single long polysaccharide molecule, as shown in Figure 6.28. Some cells are attached to the ECM by ECM glycoproteins such as **fibronectin**. Fibronectin and other ECM proteins bind to cell-surface receptor proteins called **integrins** that are built into the plasma membrane. Integrins span the membrane and bind on their cytoplasmic side to associated proteins attached to microfilaments of the cytoskeleton. The name *integrin* is based on the word *integrate*: Integrins are in a position to transmit signals between the ECM and the cytoskeleton and thus to integrate changes occurring outside and inside the cell.

Figure 6.28

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Extracellular matrix (ECM) of an animal cell.
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The structure and role of collagen fibers, proteoglycan complexes, fibronectin, and integrins are given.

The molecular composition and structure of the ECM vary from one cell type to another. In this example, three different types of ECM molecules are present: collagen, fibronectin, and proteoglycans.

Collagen fibers are embedded in a web of proteoglycan complexes. A proteoglycan complex consists of hundreds of proteoglycan molecules attached noncovalently to a single long polysaccharide molecule. Fibronectin attaches the E C M to integrins embedded in the plasma membrane. Integrins, membrane proteins with two subunits, bind to the E C M on the outside and to associated proteins attached to microfilaments on the inside. This linkage can transmit signals between the cell's external environment and its interior and can result in changes in cell behavior.

Current research on fibronectin, other ECM molecules, and integrins reveals the influential role of the ECM in the lives of cells. By communicating with a cell through integrins, the ECM can regulate a cell's behavior. For example, some cells in a developing embryo migrate along specific pathways by matching the orientation of their microfilaments to the "grain" of fibers in the extracellular matrix. Researchers have also learned that the extracellular matrix around a cell can influence the activity of genes in the nucleus. Information about the ECM probably reaches the nucleus by a combination of mechanical and chemical signaling pathways. Mechanical signaling involves fibronectin, integrins, and microfilaments of the cytoskeleton. Changes in the cytoskeleton may in turn trigger signaling pathways inside the cell, leading to changes in the set of proteins being made by the cell and therefore changes in the cell's function. In this way, the extracellular matrix of a particular tissue may help coordinate the behavior of all the cells of that tissue. Direct connections between cells also function in this coordination, as we'll explore next.

Cell Junctions

Cells in an animal or plant are organized into tissues, organs, and organ systems. Neighboring cells often adhere, interact, and communicate via sites of direct physical contact.

Plasmodesmata in Plant Cells

It might seem that the nonliving cell walls of plants would isolate plant cells from one another. But in fact, as shown in **Figure 6.29**, many plant cell walls are perforated with **plasmodesmata** (singular, *plasmodesma*; from the Greek *desma*, bond), channels that connect cells. The plasma membranes of adjacent cells line the channel of each plasmodesma and thus are continuous. Because the channels are filled with cytosol, the cells share the same internal chemical environment. By joining adjacent cells, plasmodesmata unify most of the plant into one living continuum. Water and small solutes can pass freely from cell to cell, and several experiments have shown that in some circumstances, certain proteins and RNA molecules can do this as well (see Concept 36.6). The macromolecules transported to neighboring cells appear to reach the plasmodesmata by moving along fibers of the cytoskeleton.

Figure 6.29

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Plasmodesmata between plant cells.
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Plasmodesmata visible as channels in the cells walls, connecting the cytoplasm of adjacent cells. Scale: 0.5 micrometers.

The cytoplasm of one plant cell is continuous with the cytoplasm of its neighbors via plasmodesmata, cytoplasmic channels through the cell walls (TEM).

Tight Junctions, Desmosomes, and Gap Junctions in Animal Cells

In animals, there are three main types of cell junctions: ***tight junctions****,****desmosomes***, and ***gap junctions***. (Gap junctions are most like the plasmodesmata of plants, although gap junction pores are not lined with membrane—rather, they consist of proteins extending from each cell's membrane that form a connecting pore.) All three types of cell junctions are especially common in epithelial tissue, which lines the external and internal surfaces of the body. **Figure 6.30** uses epithelial cells of the intestinal lining to illustrate these junctions.

Figure 6.30

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Exploring Cell Junctions in Animal Tissues
```

Exploring cell junctions in animal tissues.

Tight junctions: In diagram tight junctions appear as a web of proteins clamping cells together. In TEM, cells appear very tightly pressed together at a tight junction (scale: 0.5 micrometers). At tight junctions, the plasma membranes of neighboring cells are very tightly pressed against each other, bound together by specific proteins. Forming continuous seals around the cells, tight junctions establish a barrier that prevents leakage of extracellular fluid across a layer of epithelial cells. For example, tight junctions between skin cells make us water tight. Desmosomes (one type of anchoring junction) function like rivets, fastening cells together into strong sheets. Intermediate filaments made of sturdy keratin proteins anchor desmosomes in the cytoplasm. Desmosomes attach muscle cells to each other in a muscle. Some muscle tears involve the rupture of desmosomes. Gap junctions (also called communicating junctions) provide cytoplasmic channels from one cell to an adjacent cell and in this way are similar in their function to the plasmodesmata in plants. Gap junctions consist of membrane proteins that surround a pore through which ions, sugars, amino acids, and other small molecules may pass. Gap junctions are necessary for communication between cells in many types of tissues, such as heart muscle, and in animal embryos.

Animation

Cell Junctions

image

Play

image

Play

Concept Check 6.7

- In what way are the cells of plants and animals structurally different from single-celled eukaryotes?
- **WHAT IF?** If the plant cell wall or the animal extracellular matrix were impermeable, what effect would this have on cell function?
- **MAKE CONNECTIONS** The polypeptide chain that makes up a tight junction weaves back and forth through the membrane four times, with two extracellular loops and one loop plus short C-terminal and N-terminal tails in the cytoplasm. Looking at Figure 5.14, what would you predict about the amino acid sequence of the tight junction protein?

*For suggested answers, see*Appendix A.**

Concept 6.8: A cell is greater than the sum of its parts

From our panoramic view of the cell's compartmental organization to our close-up inspection of each organelle's architecture, this tour of the cell has shown the correlation of structure with function. (See Figure 6.8 to review cell structure.)

BioFlix® Animations

Tour of an Animal Cell

image

image

BioFlix® Animations

Tour of a Plant Cell

image

image

Remember that none of a cell's components work alone. For example, consider the microscopic scene in **Figure 6.31**. The large cell is a macrophage (see Figure 6.13a). It helps defend the mammalian body against infections by ingesting bacteria (the smaller cells) into phagocytic vesicles. The macrophage crawls along a surface and reaches out to the bacteria with thin pseudopodia (specifically, filopodia). Actin filaments interact with other elements of the cytoskeleton in these movements. After the macrophage engulfs the bacteria, they are destroyed by lysosomes produced by the elaborate endomembrane system. The digestive enzymes of the lysosomes and the proteins of the cytoskeleton are all made by ribosomes. And the synthesis of these proteins is programmed by genetic messages dispatched from the DNA in the nucleus. All these processes require energy, which mitochondria supply in the form of ATP.

Figure 6.31

Coordination of activities in a cell.

A macrophage has a filopodium, which is an extension of its body. The filopodium engulfs nearby bacteria. The scale is 10 micrometers.

The ability of this macrophage to recognize, apprehend, and destroy bacteria involves coordination among components such as the cytoskeleton, lysosomes, and plasma membrane (colorized SEM).

Animation

Review of Animal Cell Structure and Function

Cellular functions arise from cellular order: The cell is a living unit greater than the sum of its parts. The cell in Figure 6.31 is a good example of integration of cellular processes, seen from the outside. But what about the internal organization of a cell? As you proceed in your study of biology to consider different cellular processes, it will be helpful to try to visualize the architecture and furnishings inside a cell. **Figure 6.32** is designed to introduce you to some important biological molecules and structures and to help you get a sense of their relative sizes and organization in the context of other cellular components. As you study this figure, see if you can mentally shrink yourself down to the size of a protein and contemplate your surroundings.

Figure 6.32 Interactive Visual Activity

Visualizing the Scale of the Molecular Machinery in a Cell

Interact with this figure in the Figure 6.32 Interactive Visual Activity: Visualizing the Scale of the Molecular Machinery in a Cell.

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Advanced placement edition

Apply concepts and connect across Big Ideas using visual representations.

How do cell organelles interact to produce an enzyme? Why is ATP Synthase oriented differently in cellular respiration and photosynthesis? Which illustrated features and processes do all organisms share?

Figure 6.32

```
Visualizing the Scale of the Molecular Machinery in a Cell
```

Visualizing the scale of the molecular machinery in a cell.

A slice of a plant cell's interior is illustrated in the center panel, with selected molecules and structures shown above and below. Also summarizes cellular respiration, photosynthesis, transcription, and translation. A slice of a plant cell's interior is illustrated in the center panel, with all structures and molecules drawn to scale. Selected molecules and structures are shown above and below, all enlarged by the same factor so you can compare their sizes. All protein and nucleic acid structures are based on data from the Protein Data Bank; regions whose structure has not yet been determined are shown in gray. Information reads as follows.

- Membrane proteins: Proteins embedded in the plasma membrane or other cellular membranes help transport substances across membranes, conduct signals from one side of the membrane to the other, and participate in other crucial cellular functions. Many proteins are able to move within the membrane. Drawing shows part of the membrane with a proton pump, calcium channel, and aquaporin, which are involved in transport. Also shows a receptor and an enzyme in a signaling pathway, which are involved in signaling.
- Cellular respiration: Cellular respiration includes many steps, some carried out by individual proteins or protein complexes in the cytoplasm and the mitochondrial matrix. Other proteins and protein complexes, involved in generating A T P from food molecules, form a chain in the inner mitochondrial membrane. Drawing shows a close-up of the plasma membrane. Just outside it are two enzymes, phosphofructokinase and hexokinase. Shown inside it is the mitochondrial matrix, with complexes 1, 2, 3, and 4. Also shown are cyt C, A T P synthase, and isocitrate dehydrogenase.

- Photosynthesis: Large complexes of proteins with associated nonprotein molecules are embedded in chloroplast membranes. Together, they can trap light energy in molecules that are later used by other proteins inside the chloroplast to make sugars. This is the basis for all life on the planet. Drawing shows a floating rubisco molecule. Shown embedded in a chloroplast membrane are from left to right: photosystem 2, Pq, cytochrome complex, Pc, photosystem 1, A T P synthase, Fd, and N A D P plus reductase.

Visualizing the scale of the molecular machinery in a cell, continued.

Information continues as follows.

- Transcription: In the nucleus, the information contained in a D N A sequence is transferred to messenger R N A (m R N A) by an enzyme called R N A polymerase. After their synthesis, m R N A molecules leave the nucleus via nuclear pores. The nuclear pore complex regulates molecular traffic in and out of the nucleus, which is bounded by a double membrane. Among the largest structures that pass through the pore are the ribosomal subunits, which are built in the nucleus. Drawing shows D N A, m R N A, R N A polymerase, and a nucleosome (D N A wrapped around eight histone proteins).
- Translation: In the cytoplasm, the information in m R N A is used to assemble a polypeptide with a specific sequence of amino acids. Both transfer R N A (t R N A) molecules and a ribosome play a role. The eukaryotic ribosome, which includes a large subunit and a small subunit, is a colossal complex composed of four large ribosomal R N A (r R N A) molecules and more than 80 proteins. Through transcription and translation, the nucleotide sequence of D N A determines the amino acid sequence of a polypeptide, via the intermediary m R N A. In the middle of the page, there is a close-up of the cell nucleus pierced by a nuclear pore, surrounded by the cytoskeleton, endoplasmic reticulum, and motor proteins.
- Cytoskeleton: Cytoskeletal structures are polymers of protein subunits. Microtubules are hollow structural rods made of tubulin protein subunits, while microfilaments are cables that have two chains of actin proteins wound around each other.
- Motor proteins: Responsible for transport of vesicles and movement of organelles within the cell. This requires energy, often provided by A T P hydrolysis.

Instructors: Additional questions related to this Visualizing Figure can be assigned in **Mastering Biology**.

*For suggested answers, see*Appendix A.**

Concept Check 6.8

- *Colpidium colpoda* is a unicellular protist that lives in freshwater, eats bacteria, and moves by cilia (see Figure 6.23b). Describe how the parts of this cell work together in the functioning of *C. colpoda*, including as many organelles and other cell structures as you can.

*For suggested answers, see*Appendix A.**

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